

# ECEN721: Optical Interconnects Circuits and Systems

## Spring 2026

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### Lecture 13: CMOS Silicon Photonic PAM4 Transceiver Front-End Circuits



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Texas A&M University

# Outline

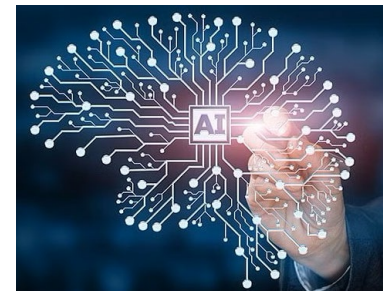
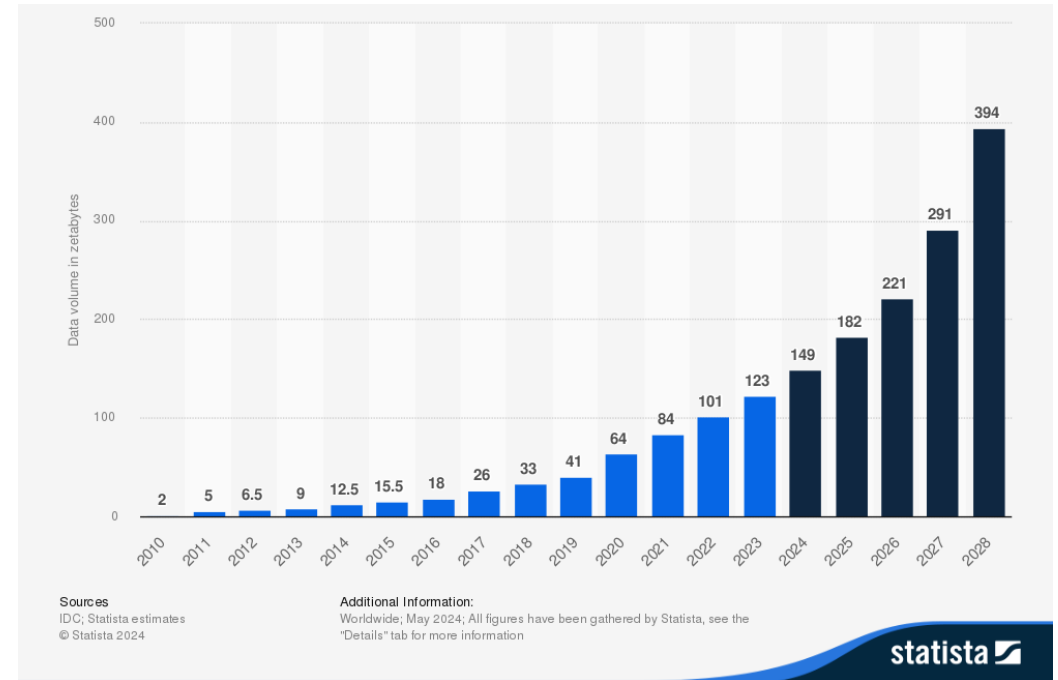
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- Motivation
- 80Gb/s PAM4 Optical DAC Microring Modulator Transmitter
- 100Gb/s PAM4 Optical Receiver with All-Silicon Microring APD
- 100Gb/s 1.32pJ/b PAM4 Optical Receiver with Digital CDR
- Conclusion

# Exponentially Increasing Data Demands

- Paradigm shift in data generation and processing
- Increased data generation
  - Artificial intelligence
  - Remote work/entertainment
  - Mobile communication
- Growing machine-to-machine traffic
- Hyperscale data centers are being optimized to infer intelligence from large data sets

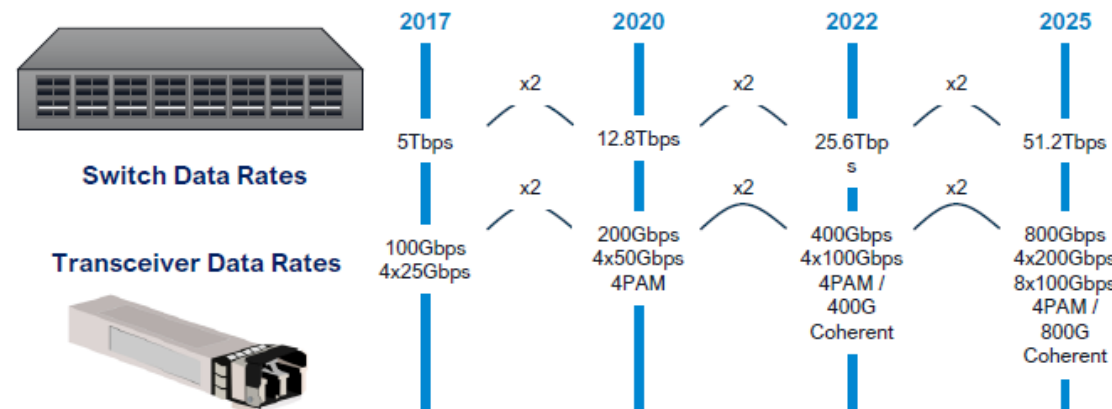
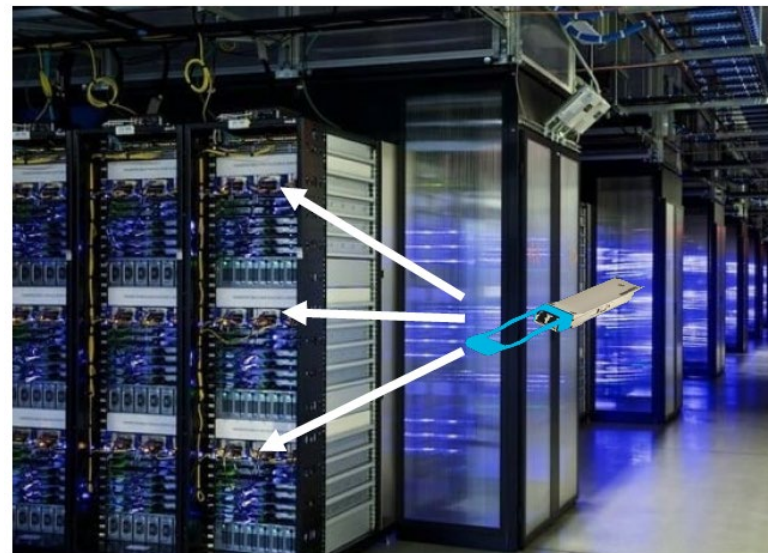
## Worldwide Data Volume (Zettabytes)



# AI Datacenter Connectivity Demands

- Scaling up with more xPUs in compute clusters
- Scaling out to larger clusters
- Higher per-lane data rates
- Flatter network topologies with higher radix switches to reduce latency
- Pushing optical connectivity to within the rack

[Wang ISSCC 2024]



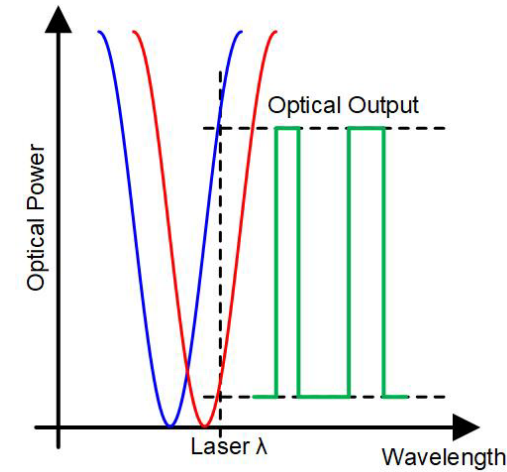
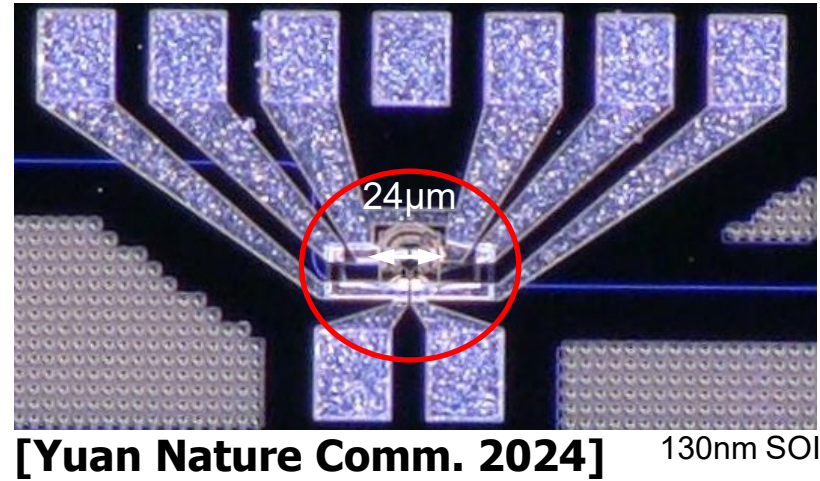
[Chan Carusone ISSCC 2024] 4



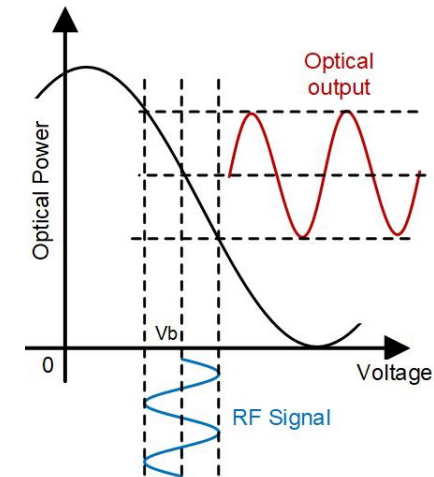
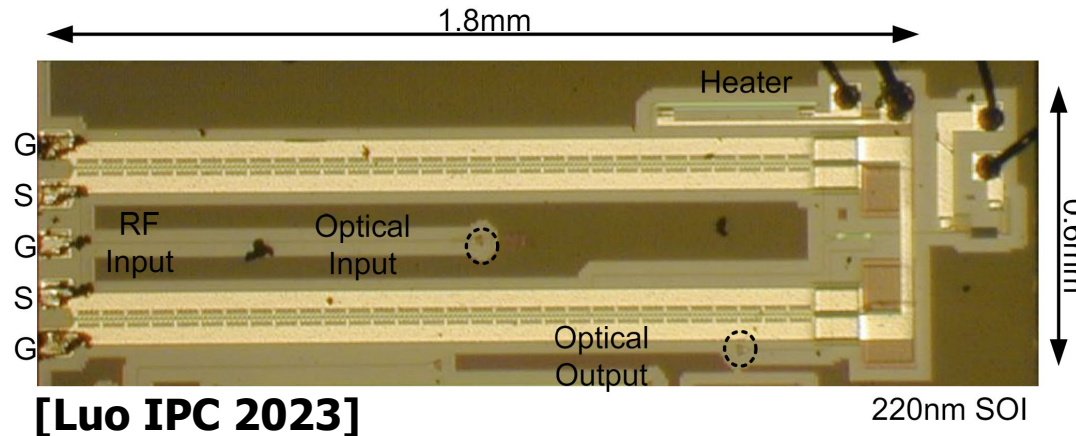
# Microring Resonator Modulator (MRM)

- External CW laser input required
- Smaller size ( $\mu\text{m}$ -scale) than Mach-Zehnder modulator (MZM) (mm-scale)
- Lower device capacitance and higher bandwidth
- Inherent WDM capability in MRM

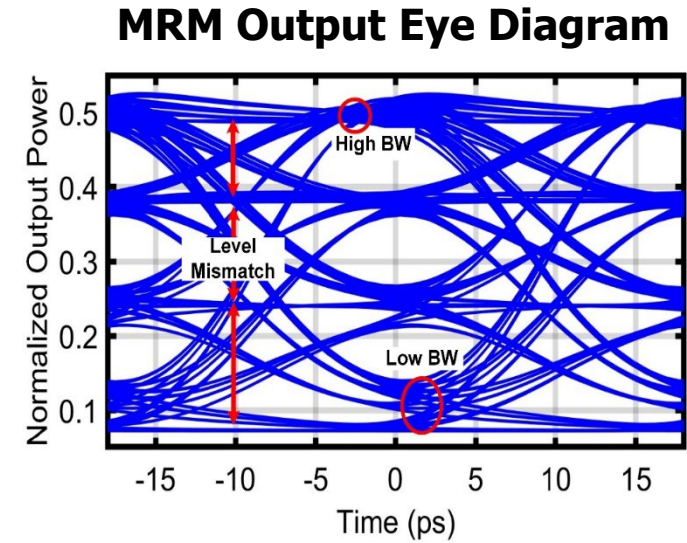
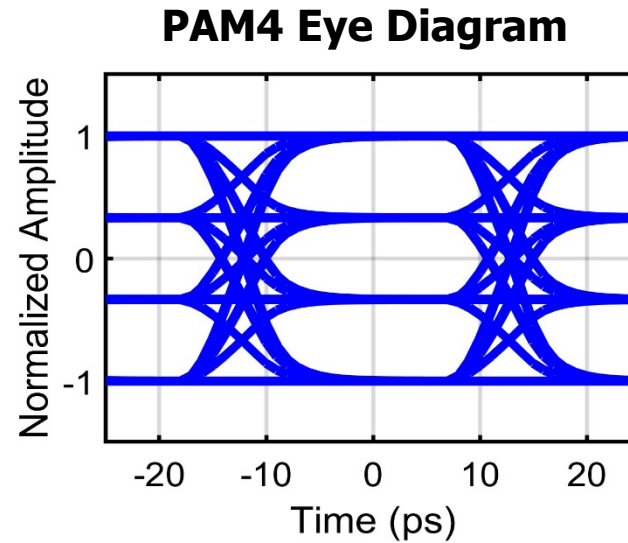
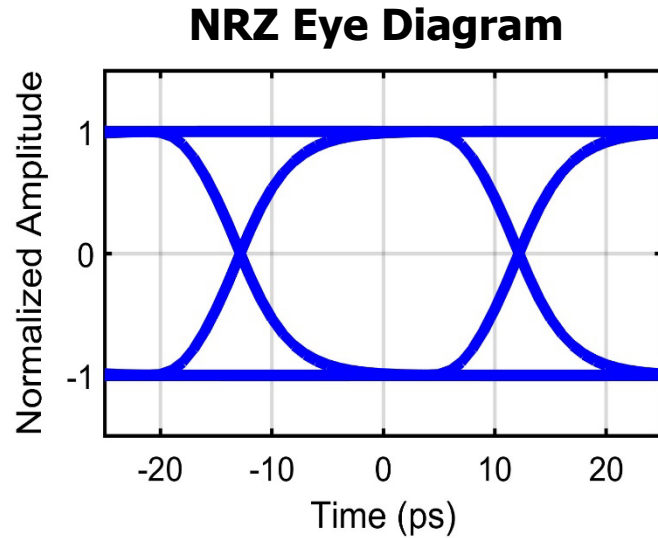
**MRM**



**MZM**



# PAM4 Modulation



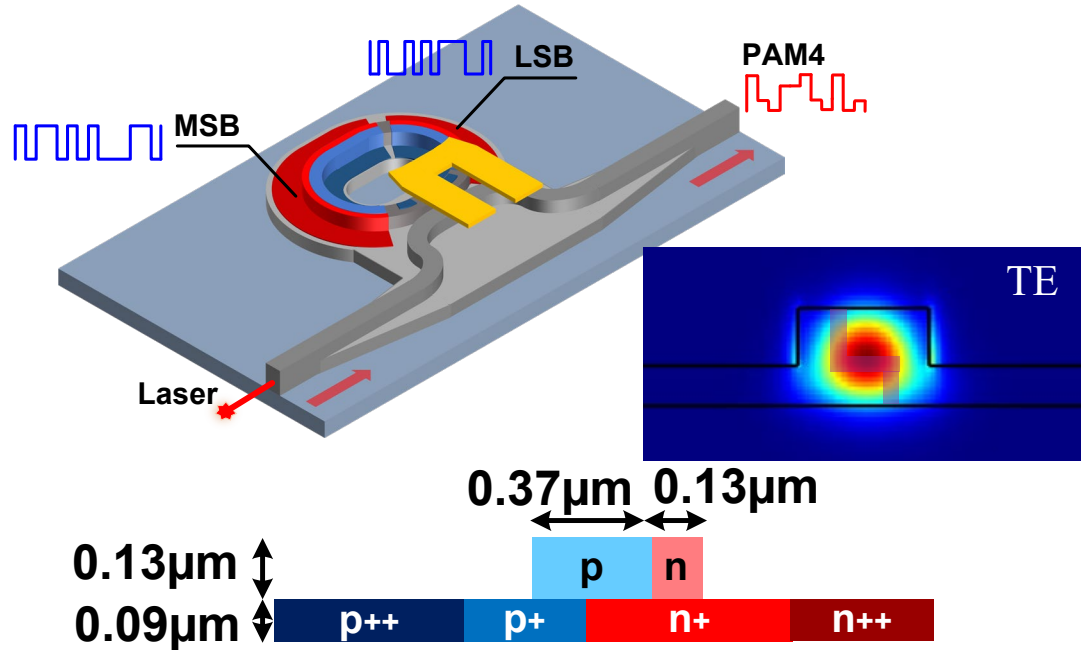
- Combines two bits into a single symbol with four levels
- Cost-effective solution for 400G/800G short-haul communication
- Easily combined with MRM, provided that static/dynamic non-linearities are addressed

# Outline

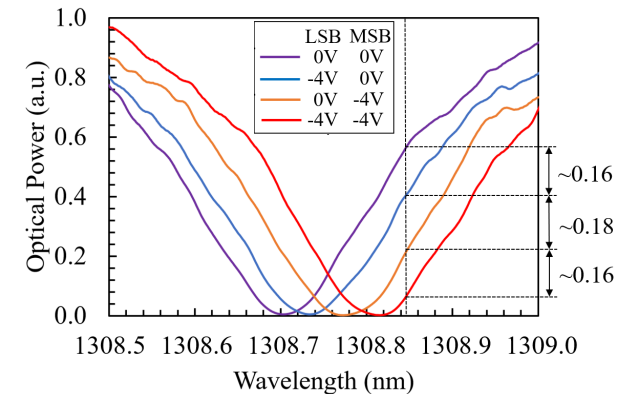
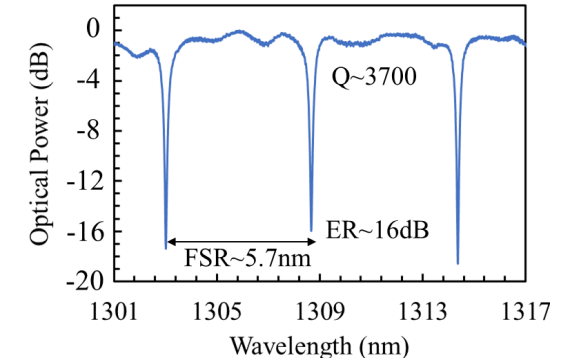
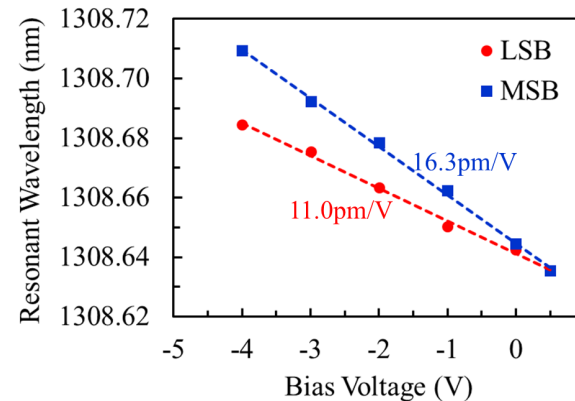
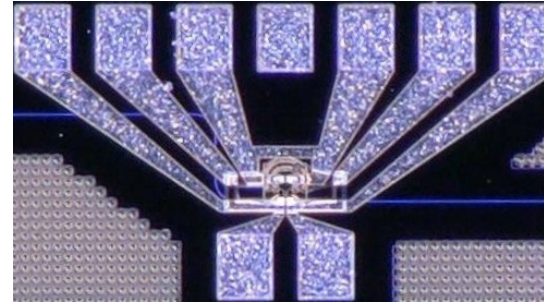
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# Two-Segment Microring Modulator



[Yuan Nature Comm. 2024]

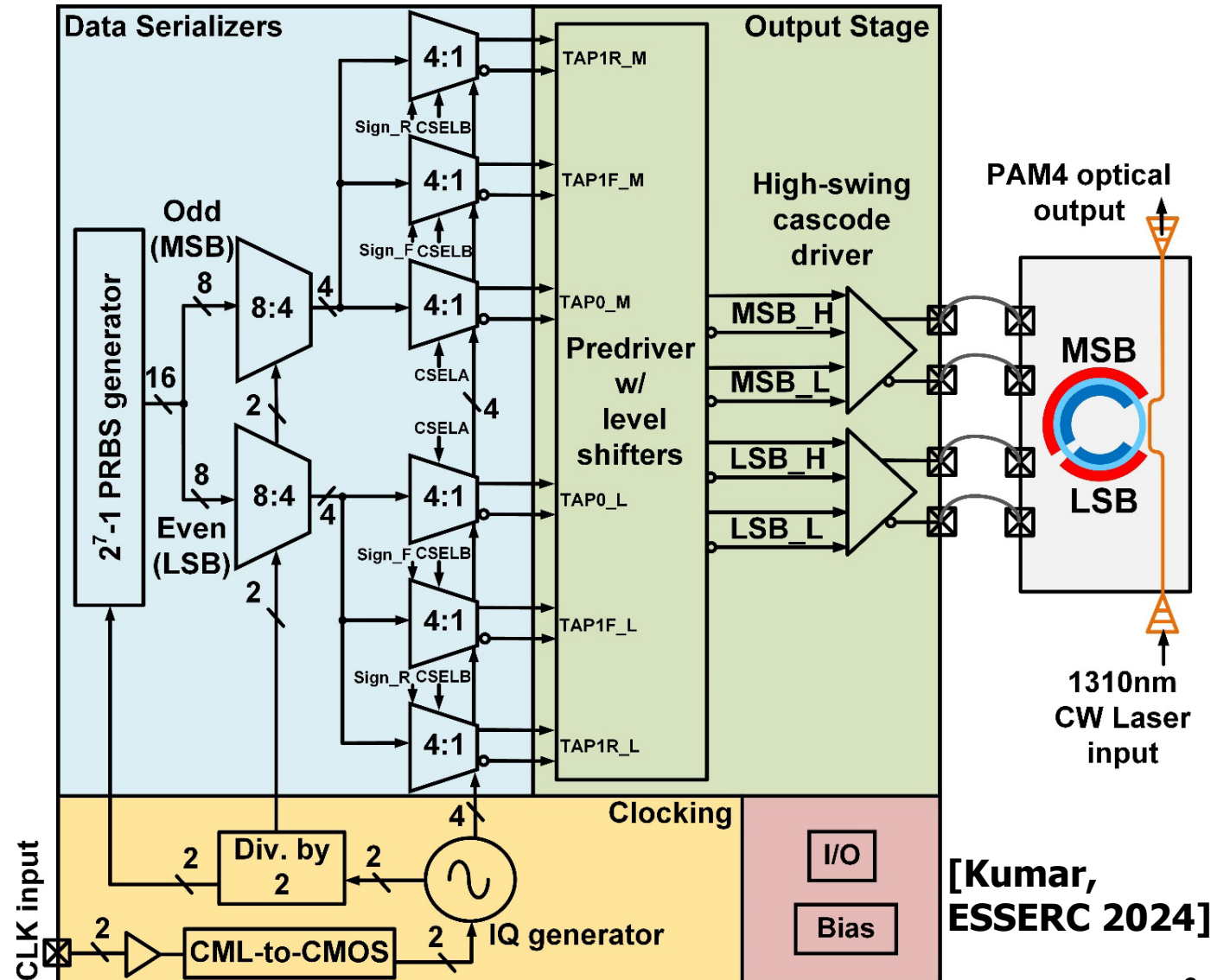


- Device length segmented with 2:1 (MSB:LSB) ratio
- Z-shaped profile improves bandwidth
- Small tunability ( $\sim 27\text{pm/V}$ ) necessitates high-swing driver



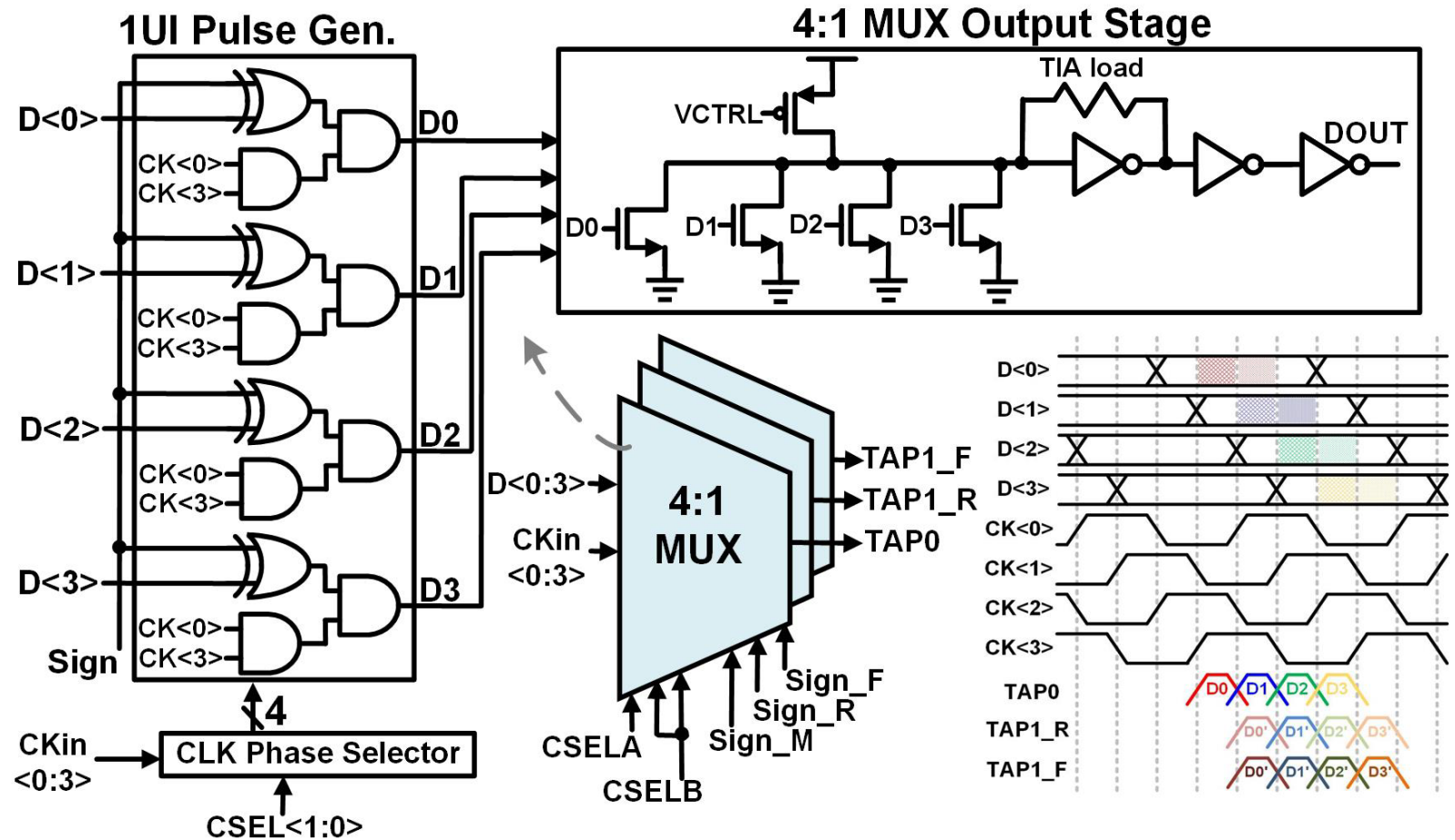
# PAM4 Transmitter Architecture

- MSB/LSB PAM2 inputs to the MRM gives optical PAM4 output
- Quarter-rate architecture with final 4:1 serialization stages
- Independently tunable high-swing cascode output drivers



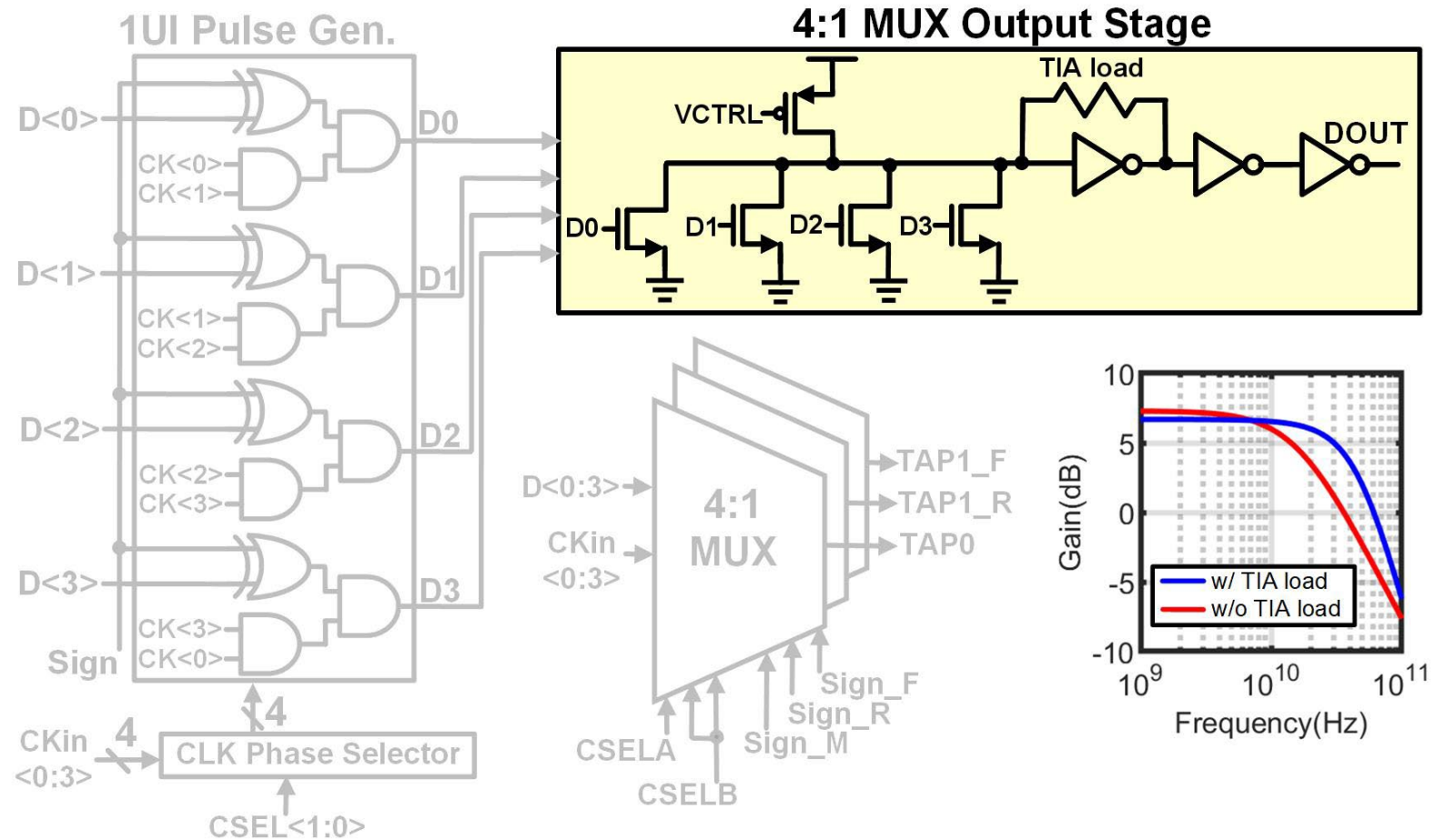
# 4:1 Serializer

- Final stage of data serialization
  - Even/Odd bits from 16-bit PRBS generator serialized to  $1/4^{\text{th}}$  rate to provide MSB/LSB streams
- Three separate 4:1 MUX segments to implement 1-tap asymmetric FFE
- Pulse generator carves 1UI pulses from quarter-rate data and clocks



# 4:1 Serializer

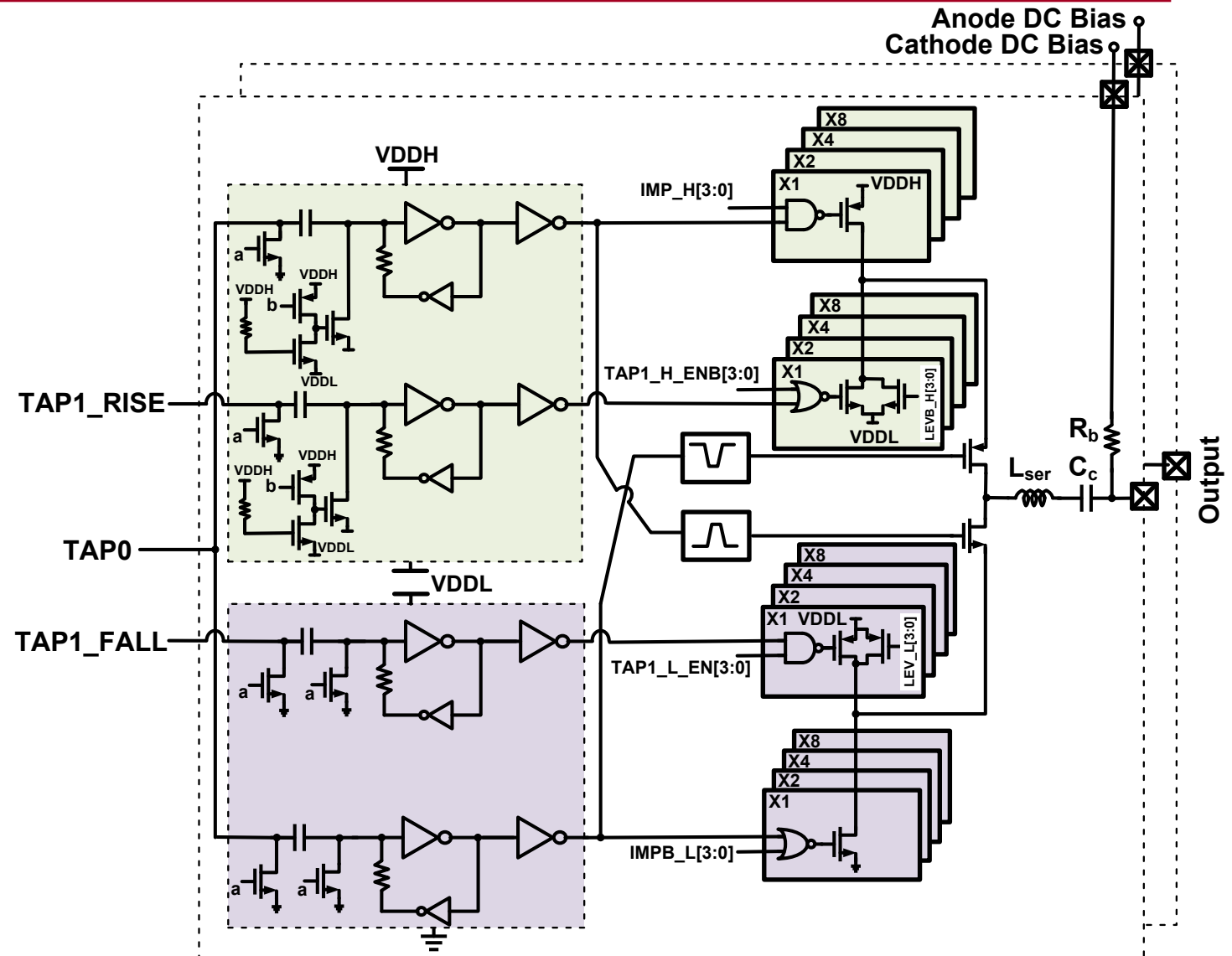
- Tailless MUX followed by TIA/buffer stages produce full-rate CMOS-level outputs
  - TIA load extends bandwidth by 2.3x at the serializer output





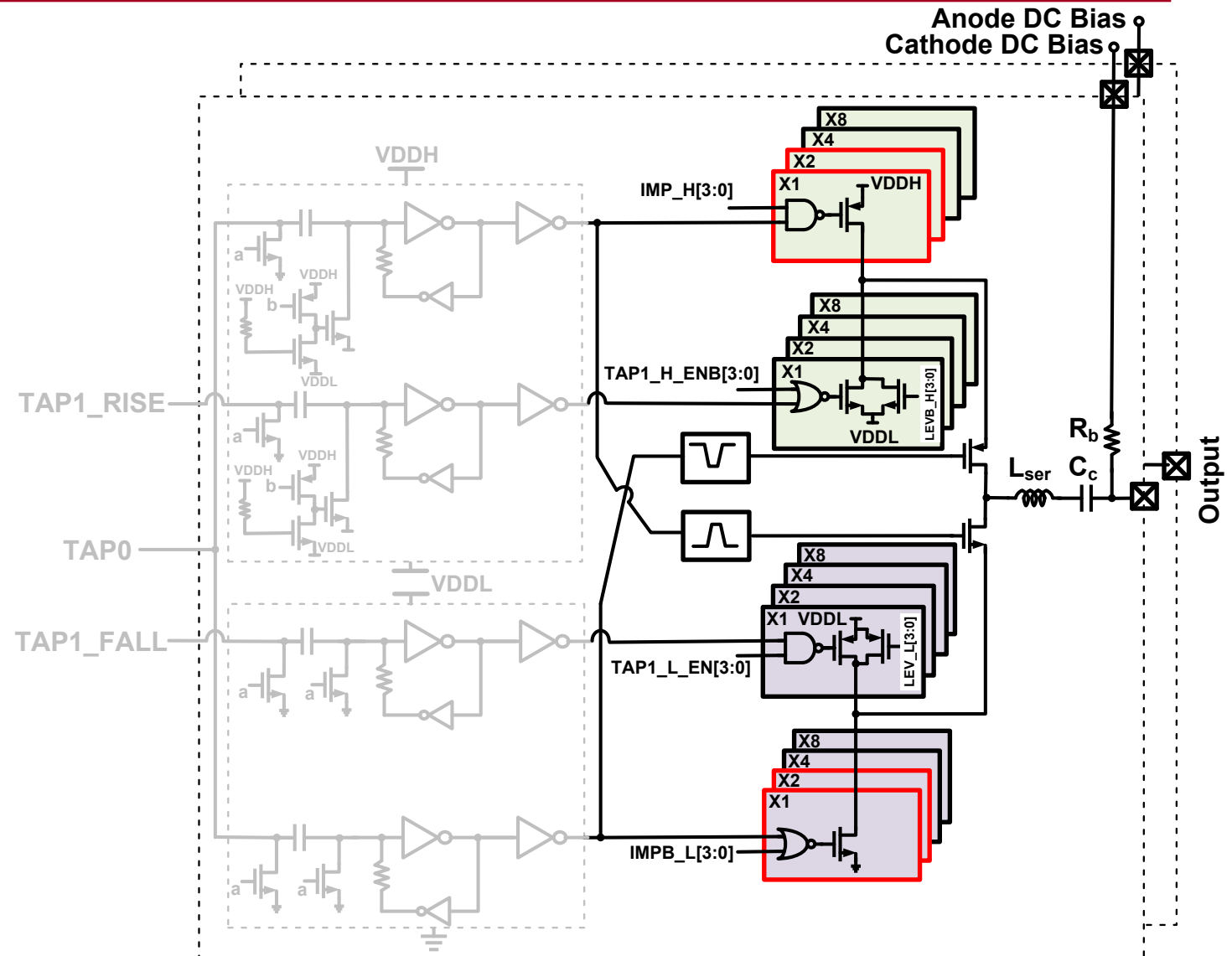
# Output Stage

- High-swing ( $\sim 4X V_{dd}$ ) differential pulsed cascode driver
- Level-shifted pre-driver provides input data for higher voltage domain in the cascode stage
- Segmented output stage provides edge rate (skew) control, output swing control, and asymmetric FFE



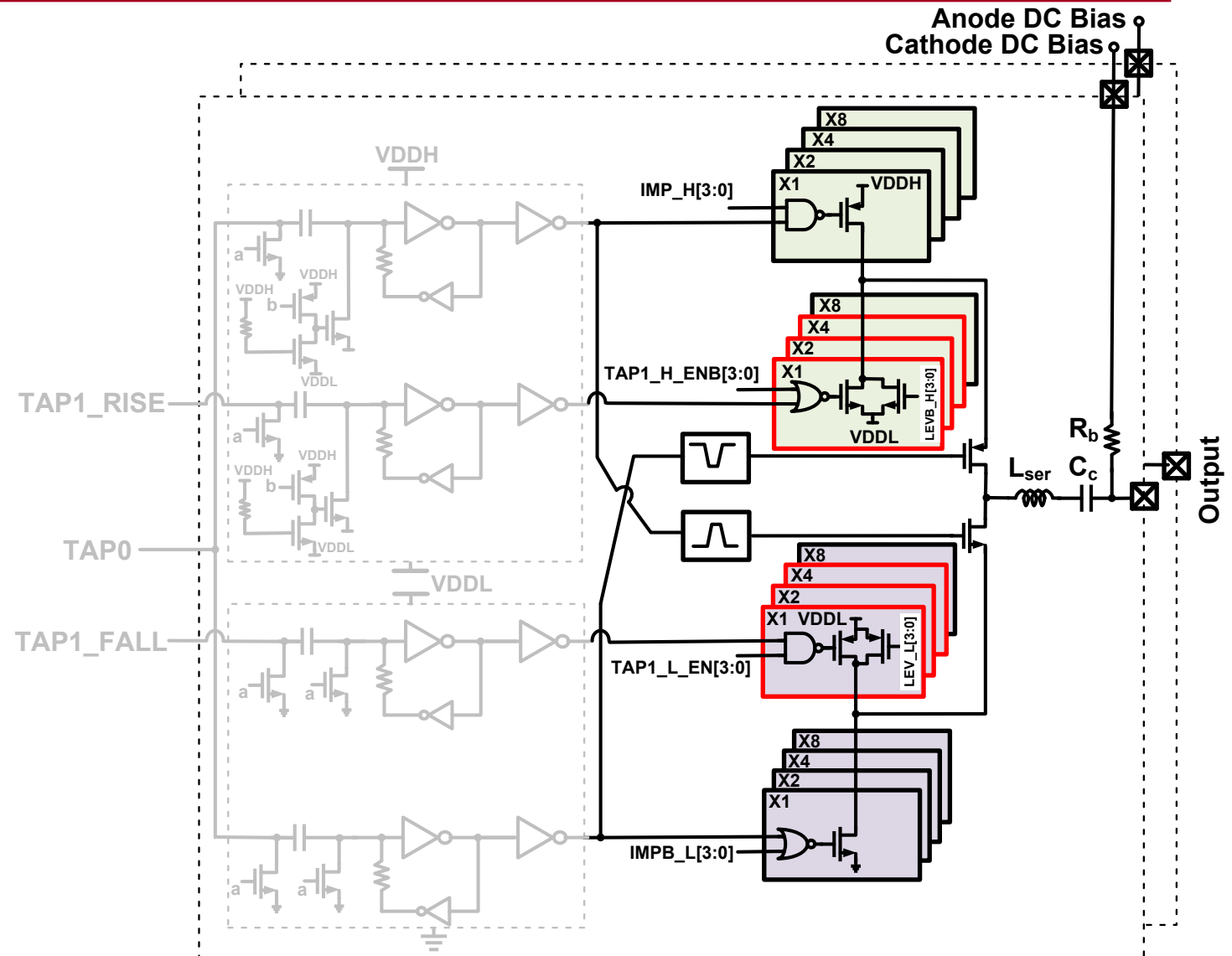
# Output Driver

- MSB/LSB segmented drivers
  - Independent edge rate control for MSB/LSB
  - Voltage level and swing control
  - Post-cursor tap implements asymmetric FFE
- Pulsed-cascode
  - Prevents  $V_{ds}$  overstress
  - All transitions are pulsed



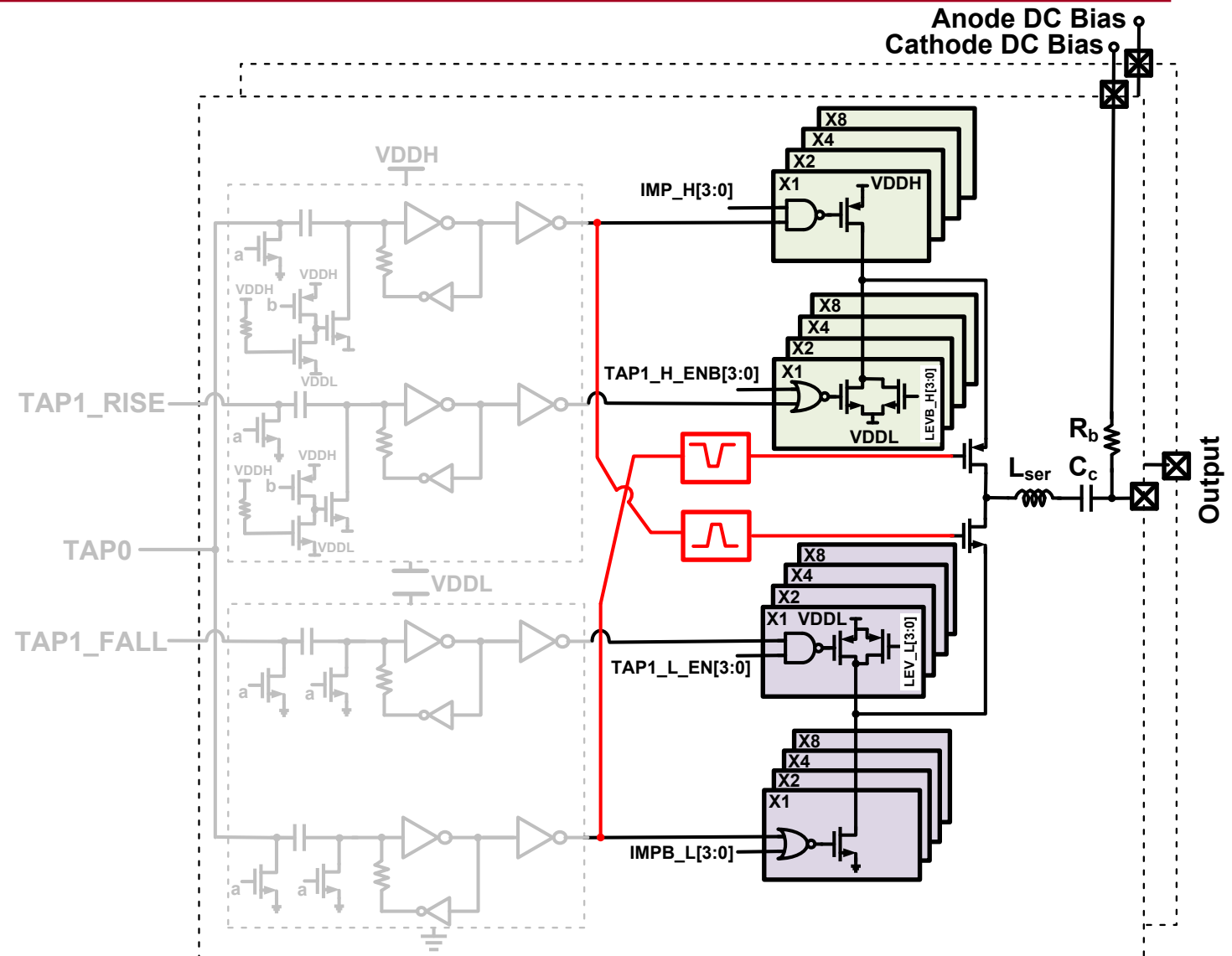
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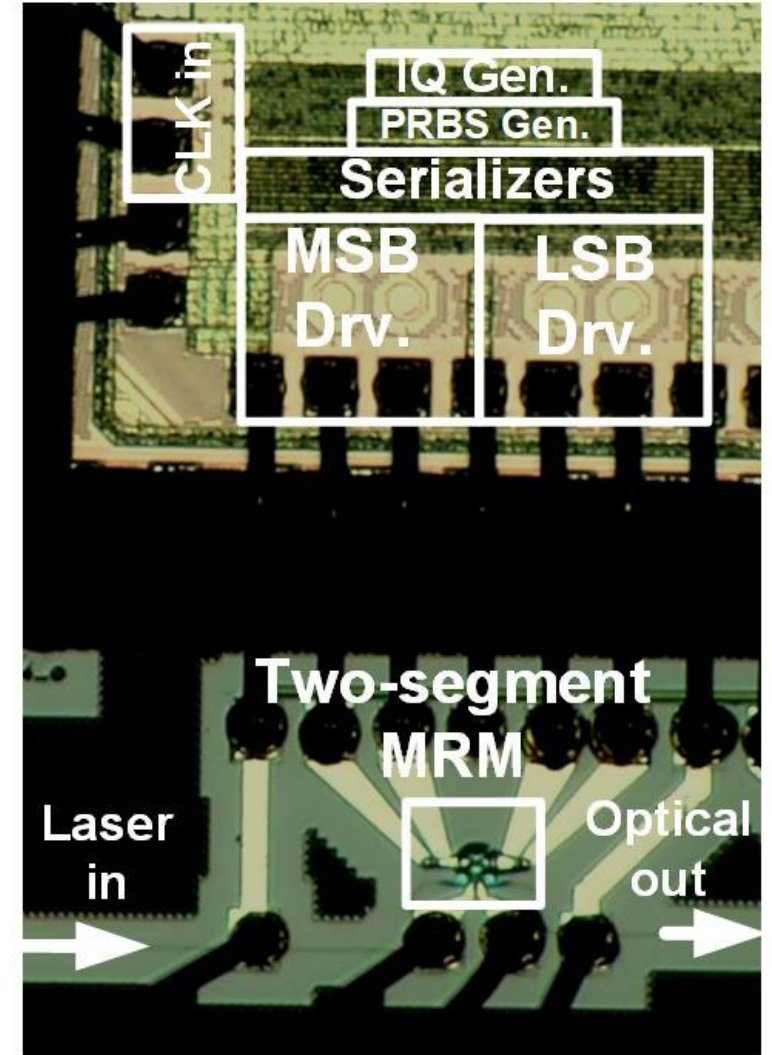
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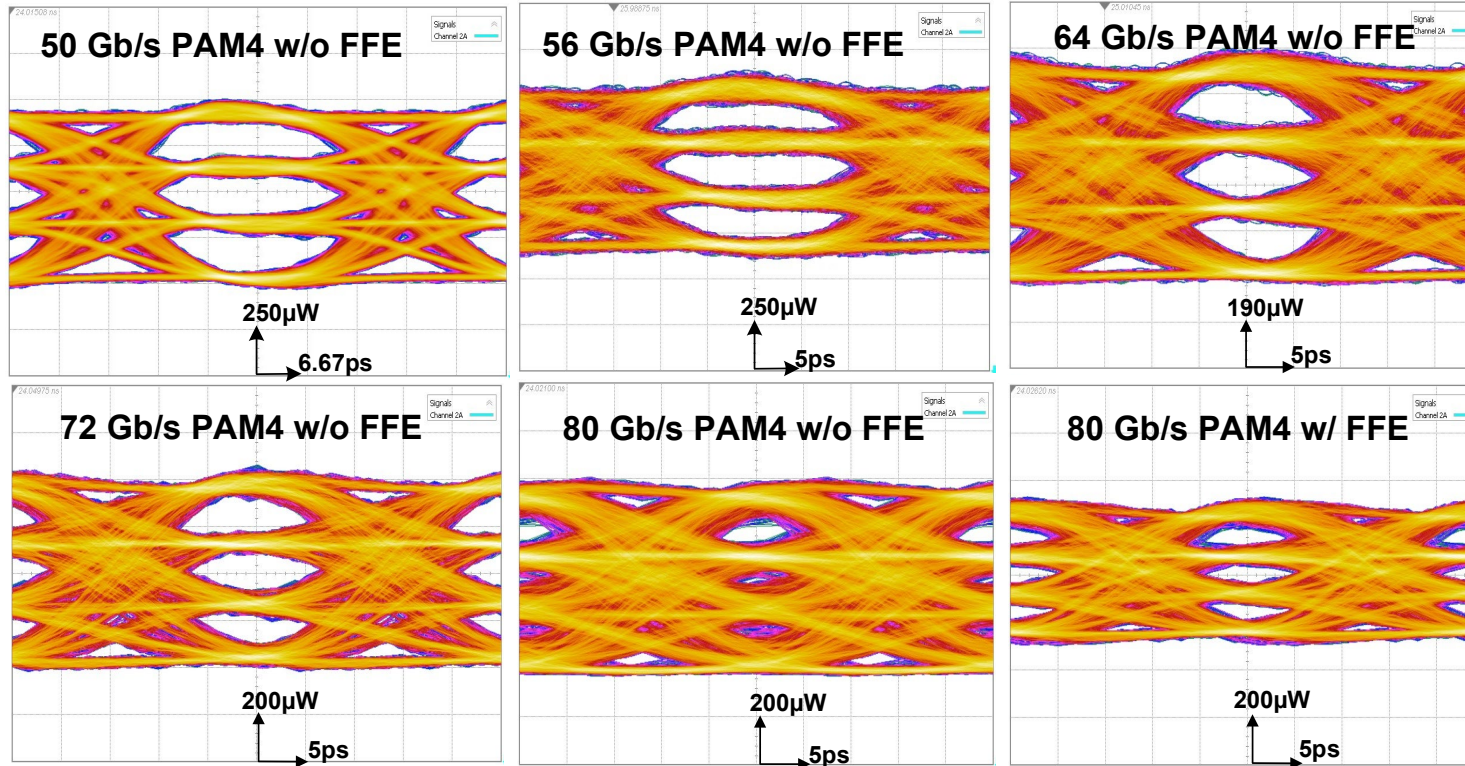
# Heterogeneous Integration

- CMOS chip
  - 28nm bulk process
- Ring-resonator chip
  - 130nm SOI process
- Integrated using wire-bonding
- Input/Output fiber vertically coupled using grating couplers





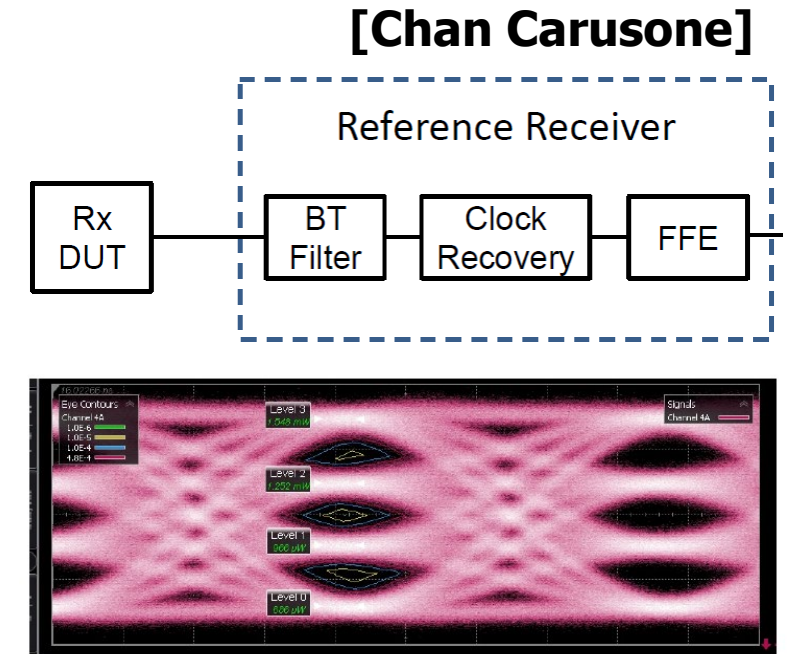
# TX Measurement Results



- Measured PAM4 optical eyes after applying level-control and edge-rate control
- 80Gb/s PAM4 eyes obtained after enabling FFE

# Transmitter and Dispersion Eye Closure Quaternary (TDECQ)

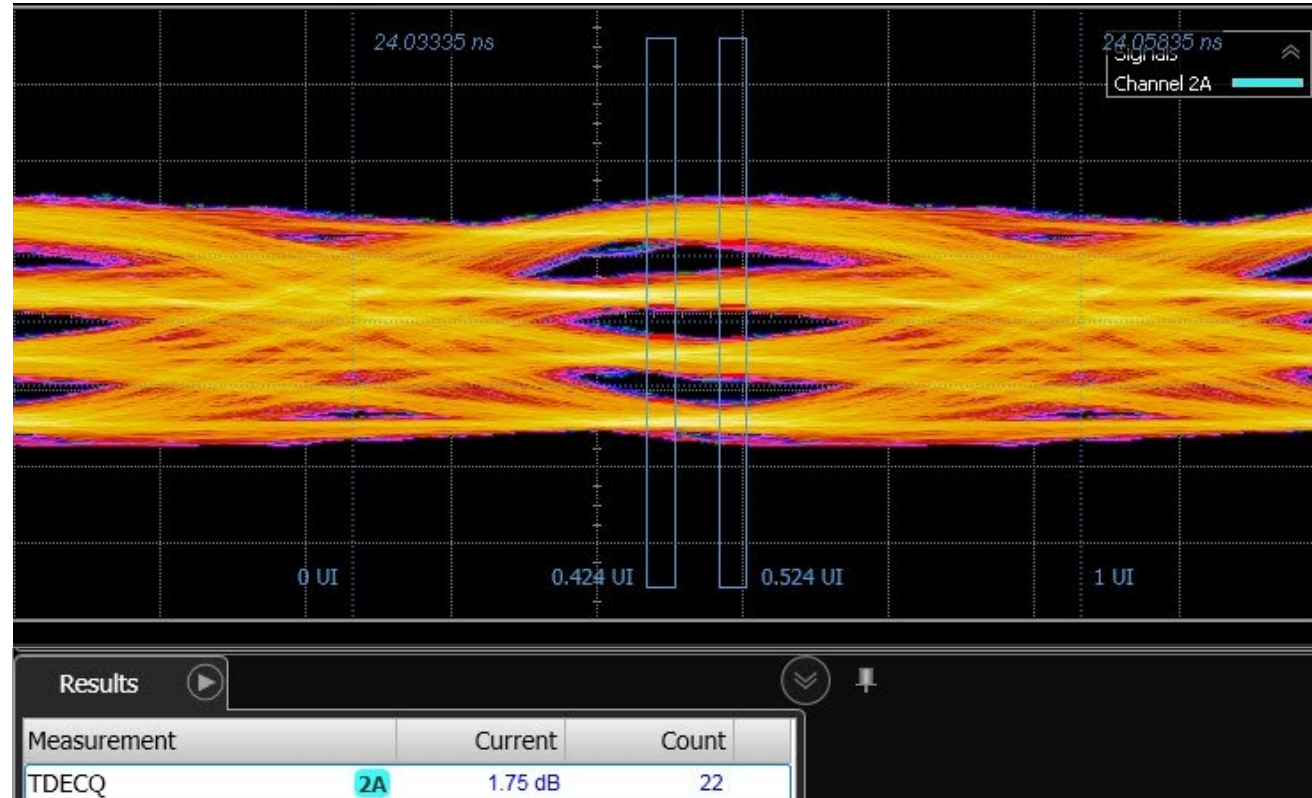
- Optical transmitter specification that captures the impact of TX nonidealities on RX OMA sensitivity
  - TX with 3dB TDECQ will require the RX OMA sensitivity to be 3dB better for the same SER
- Measured with a reference receiver consisting of a 4<sup>th</sup>-order Bessell-Thomson lowpass filter and 5-tap FFE
- Ideal PAM4 tolerable noise:  $\sigma_{ideal} = \frac{OMA}{2Q^{-1}(\frac{2}{3}SER_{target})}$
- Measurement calculates the actual noise that can be added to the TX to achieve the target SER, which will be less than  $\sigma_{ideal}$  due to nonidealities



$$TDECQ = 10 \cdot \log_{10}\left(\frac{\sigma_{ideal}}{\sigma_R}\right)$$

- Typical spec: TDECQ < 2.4dB

# TDECQ Measurement



- Measured transmitter dispersion eye closure quaternary (TDECQ) of 1.75dB at 80Gb/s PAM4 data-rate

# Transmitter Performance Comparison

|                                              | M. Moralis-Pegios<br>Optica 2020 | F. Logan<br>IPC 2020 | H. Li<br>JSSC 2021 | J. Sharma<br>JSSC 2022 | C. Levy<br>JSSC 2024 | This Work       |                                     |
|----------------------------------------------|----------------------------------|----------------------|--------------------|------------------------|----------------------|-----------------|-------------------------------------|
| <b>Modulator type</b>                        | Single-segment                   | Two-segment          | Single-segment     | Single-segment         | Single-segment       | Two-segment     |                                     |
| <b>EIC process</b>                           | 55nm SiGe BiCMOS                 | 22nm CMOS            | 28nm CMOS          | 28nm CMOS              | 28nm CMOS            | 28nm CMOS       |                                     |
| <b>Modulation</b>                            | NRZ                              | PAM4                 | PAM4               | PAM4                   | NRZ                  | PAM4            |                                     |
| <b>Data-rate (Gb/s)</b>                      | 50                               | 100                  | 112                | 112                    | 50                   | 80              |                                     |
| <b>TX architecture</b>                       | -                                | -                    | Half-rate          | Half-rate              | Quarter-rate         | Quarter-rate    |                                     |
| <b>Integration</b>                           | Wire-bond                        | Cu-Pillar            | Cu-Pillar          | Wire-bond              | Flip-chip            | Wire-bond       |                                     |
| <b>Driver output swing (V<sub>ppd</sub>)</b> | 2                                | 3.2                  | 3                  | 3                      | 1.8*                 | 3.42            |                                     |
| <b>Outer extinction ratio (dB)</b>           | 3.17                             | 5                    | 2.23               | 2.7                    | 5.3                  | 5.41            |                                     |
| <b>Energy efficiency (pJ/b)</b>              | 2 <sup>#</sup>                   | 2.5 <sup>#</sup>     | 6                  | 5.8                    | 2.5                  | 3.66            |                                     |
| <b>Transmitter FOM</b>                       | 1.0 pJ/(bit·V)                   | 0.78 pJ/(bit·V)      | 2 pJ/(bit·V)       | 1.93 pJ/(bit·V)        | 1.38 pJ/(bit·V)      | 1.07 pJ/(bit·V) | 0.79 pJ/(bit·V)<br>(Data-path only) |

<sup>#</sup>EIC datapath, no clock; \*Single-ended swing; Transmitter FOM = Power/(Data-Rate · Swing)

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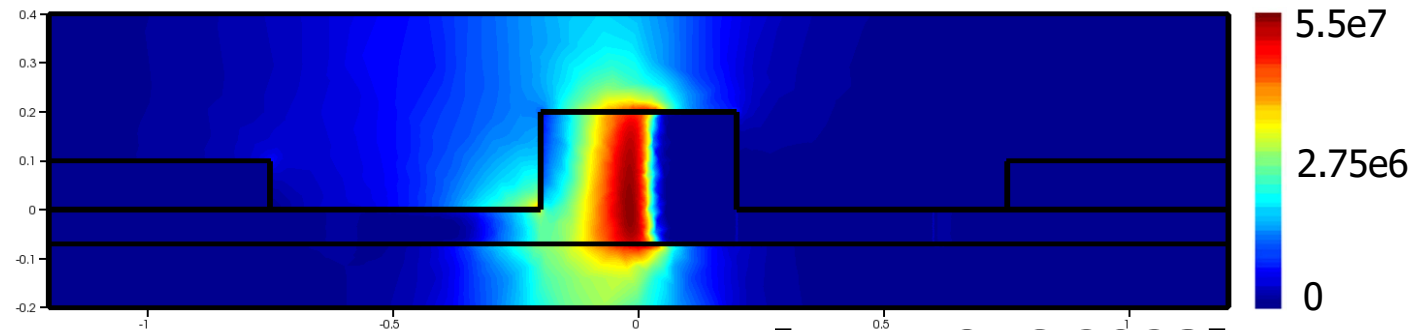
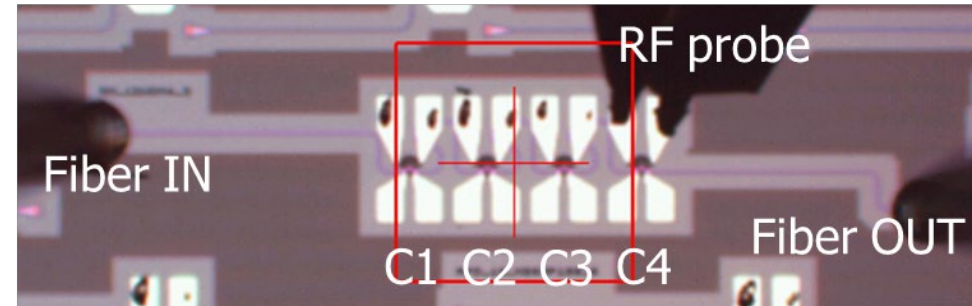
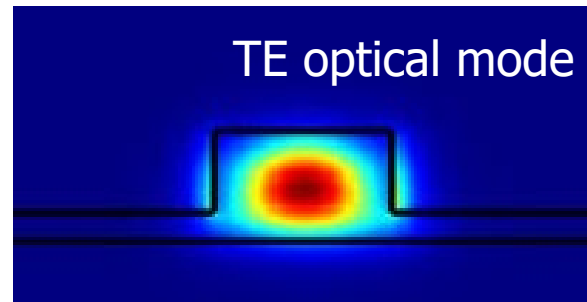
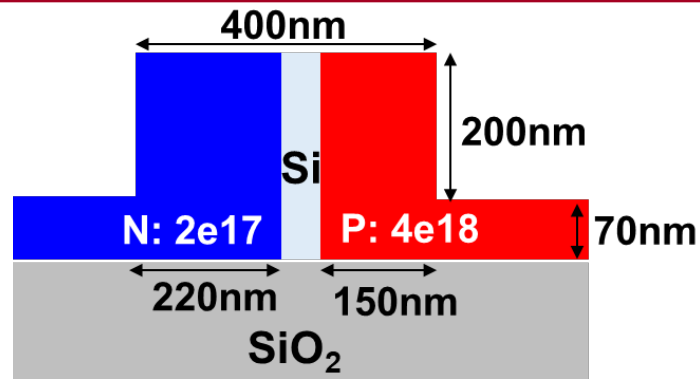
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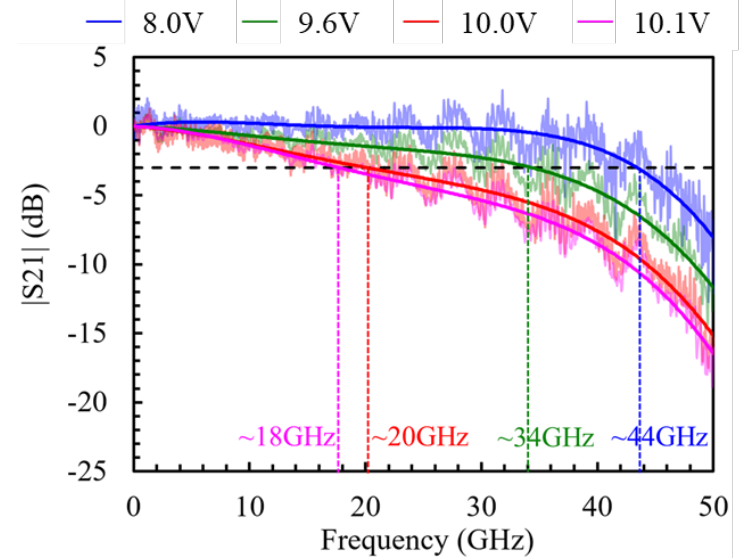
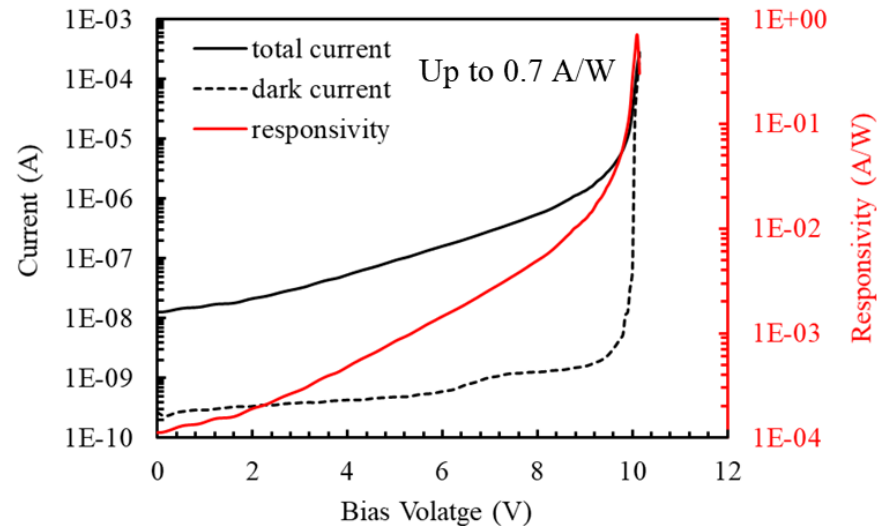
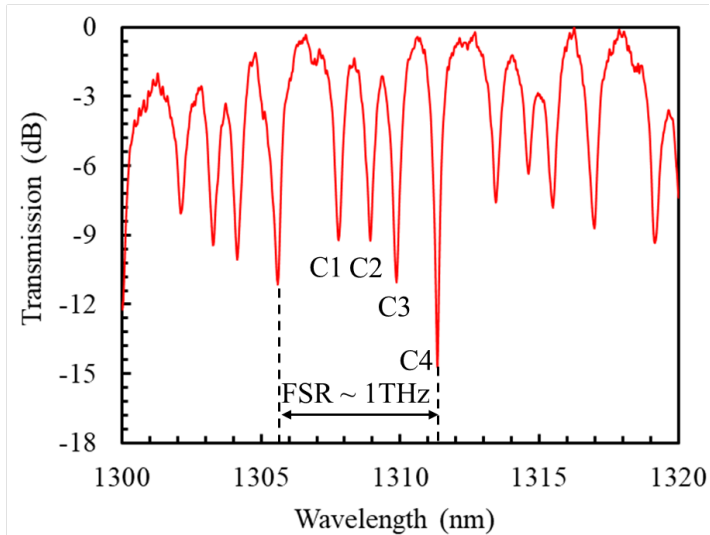
# All-Silicon Microring APD Design



[Yuan OFC 2023]

- Asymmetric junction design to place the depletion region in the center of the waveguide to maximize the overlap with the TE optical mode

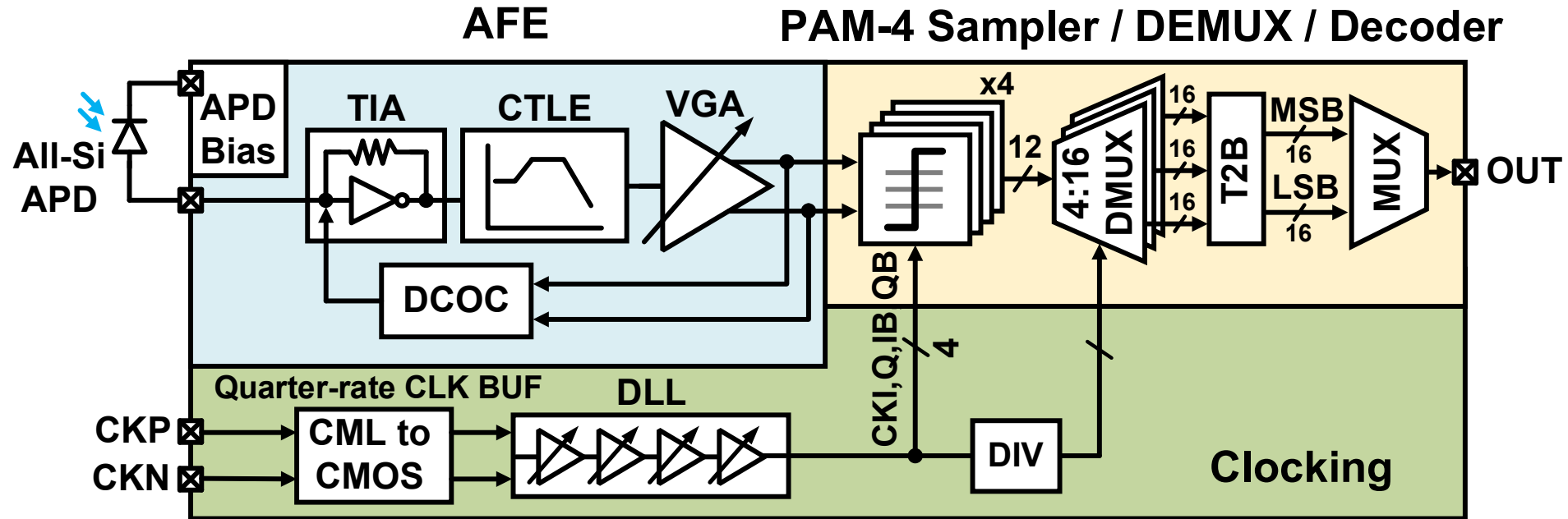
# All-Silicon Microring APD



**[Yuan OFC 2023]**

- $\sim 1$ THz FSR
- 0.7 A/W max responsivity
- Highest bandwidth of 44GHz at -8V

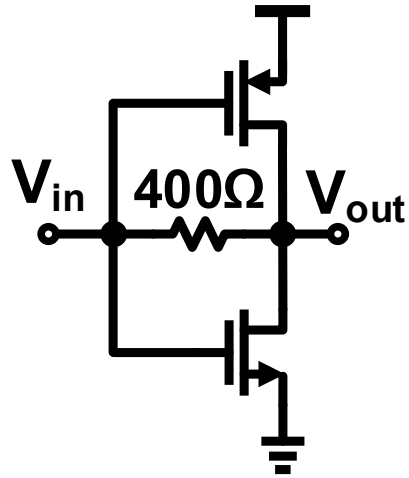
# Optical Receiver Architecture



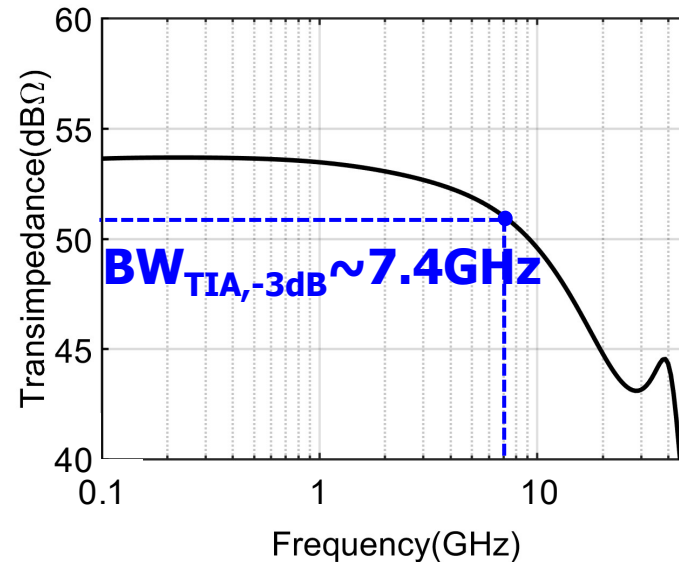
- All-Silicon APD integrated with CMOS RX through wire-bonding
- Low-noise AFE design by combining sub-Nyquist TIA with BW recovering CTLE/VGA
- Quarter-rate architecture lowers clock frequency and relaxes timing constraint

# Transimpedance Amplifier

**TIA Schematic**



**TIA Frequency Response**

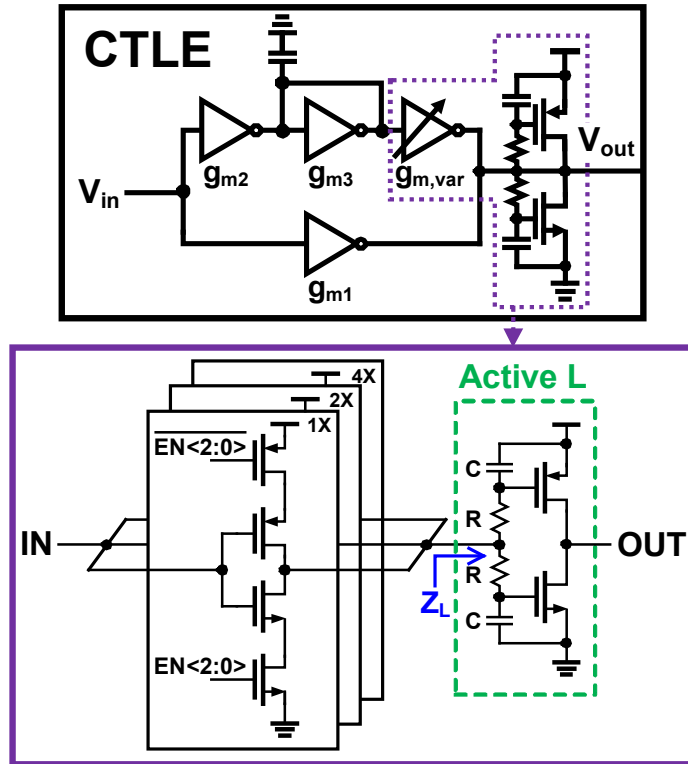


**TIA BW vs Input-Ref. Noise**

|                              | This work               | Comparison            |
|------------------------------|-------------------------|-----------------------|
| <b>TIA BW<sub>3-dB</sub></b> | 7.4GHz<br>(Sub-Nyquist) | 33.4GHz<br>(2/3*Baud) |
| <b>Input-Ref. Noise</b>      | 2.32μA <sub>rms</sub>   | 4.06μA <sub>rms</sub> |

- Sub-Nyquist TIA BW for input-referred noise improvement
- Relative to broadband TIA ( $BW_{3-dB} = 2/3 \cdot \text{Baudrate}$ ), 43% improvement in input-referred noise

# Continuous-Time Linear Equalizer

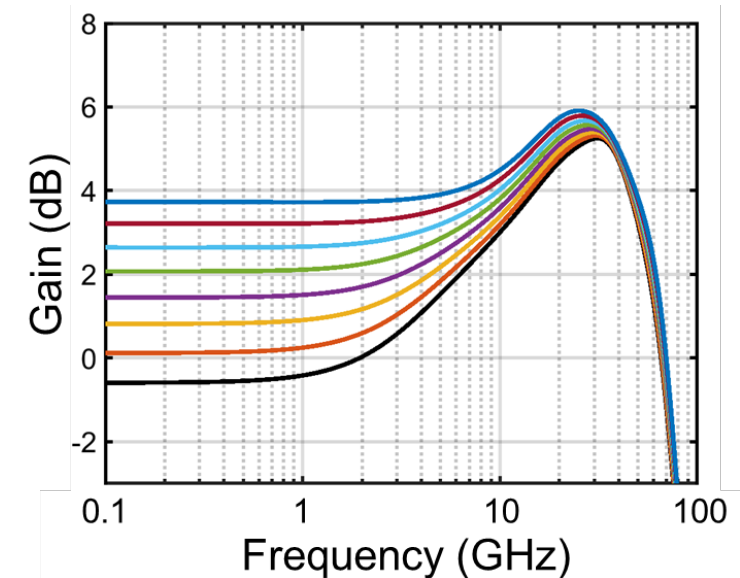


## Transfer Function of CTLE

$$H(s) = - \left( g_{m1} - g_{m,var} \frac{g_{m2}/g_{m3}}{1 + sC/g_{m3}} \right) Z_L$$

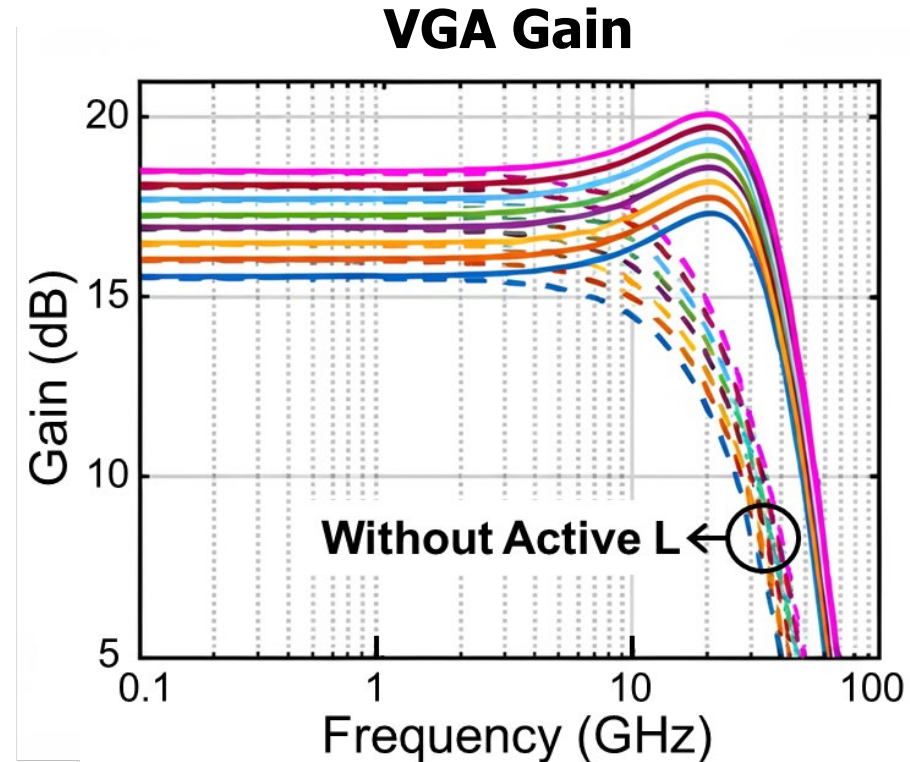
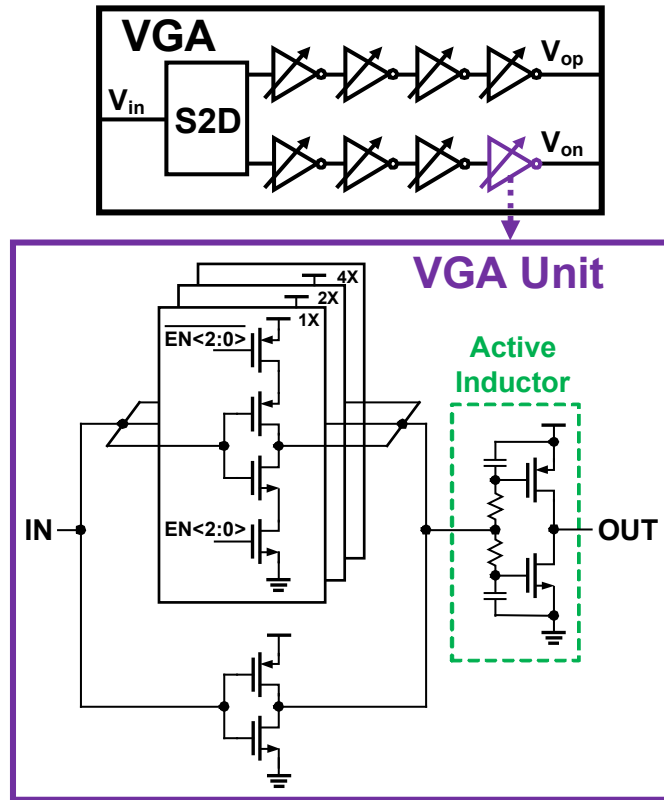
$$Z_L \approx \frac{1}{g_m} \left( 1 + \frac{s}{\omega_z} \right) = \frac{1}{g_m} + s \frac{R}{\omega'_T}$$

## CTLE Gain



- CTLE employs an active inductor to recover bandwidth, introducing up to 5.5dB of peaking

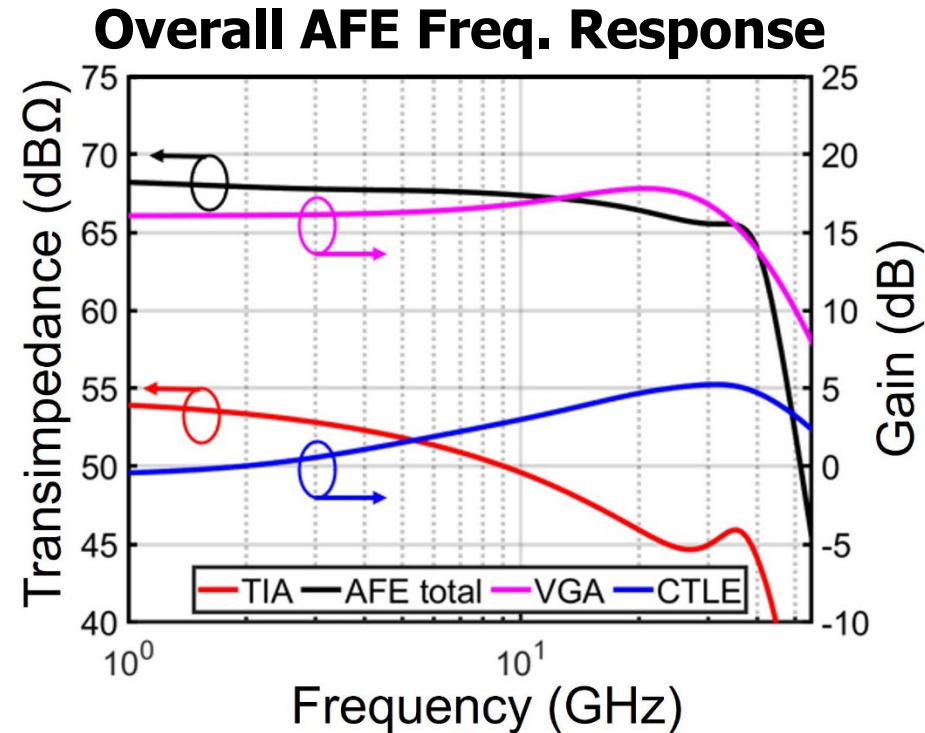
# Variable Gain Amplifier



- Gain adjustment through transconductance control
- Active inductor provides peaking, extending VGA BW



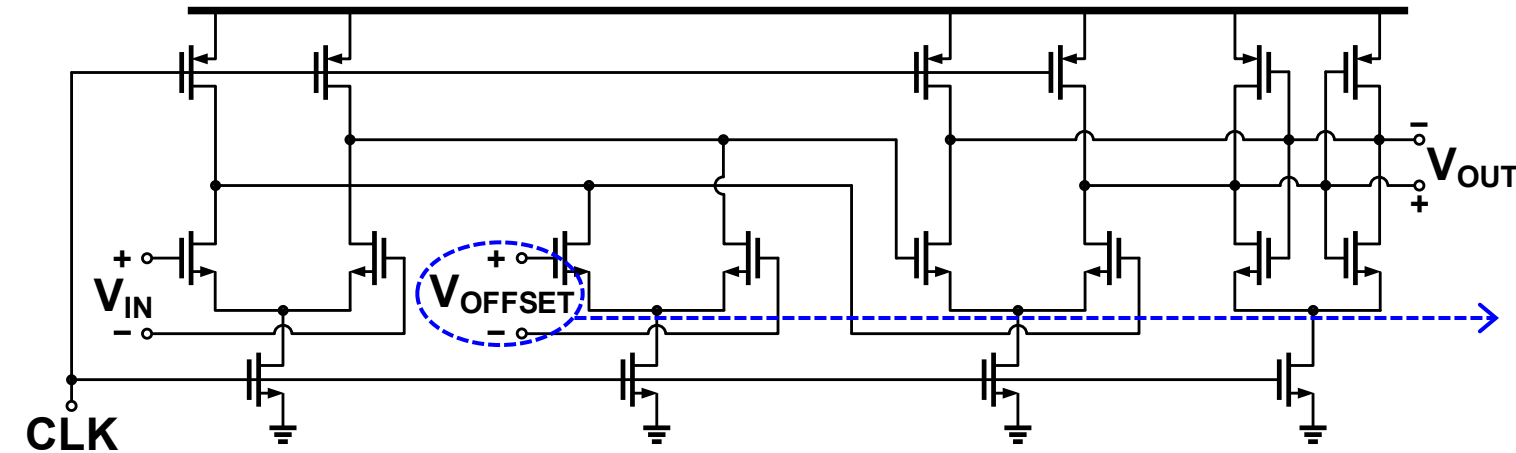
# Overall AFE Frequency Response



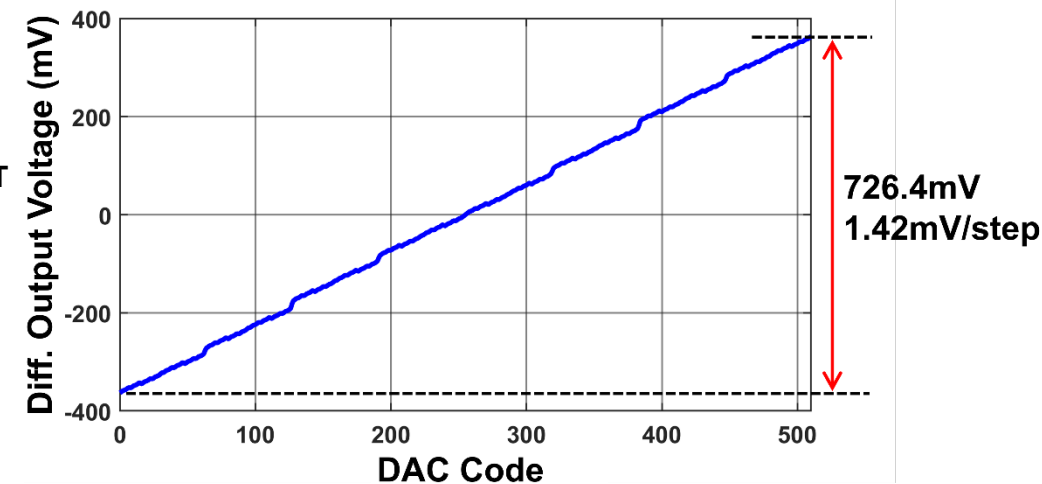
- TIA bandwidth is extended  $\sim 5\times$  by CTLE and VGA for an overall AFE bandwidth around 38GHz

# PAM4 Samplers

## Sampler Schematic



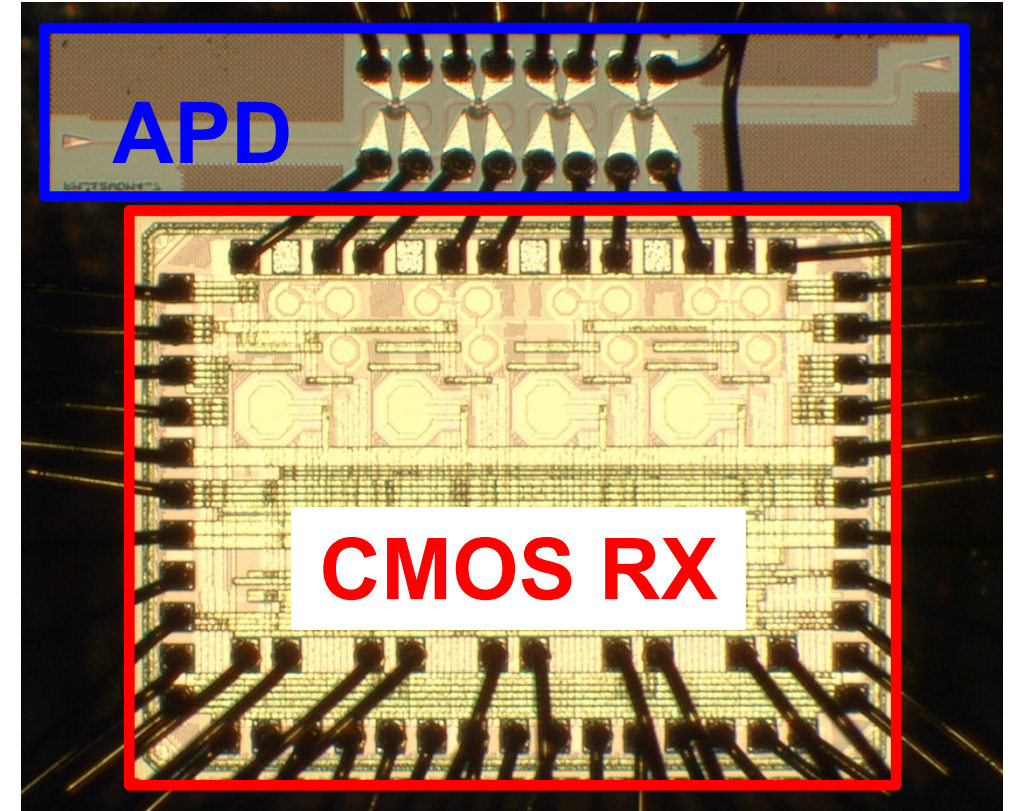
## DAC Output Voltage



- PAM4 signal digitized by 4 quarter-rate groups of 3 samplers and subsequent thermo-to-binary decoder
- 8-bit DACs generate offset/threshold control voltage
- Only 0.1dB power penalty due to finite DAC resolution ( $1.4e-4$  BER)

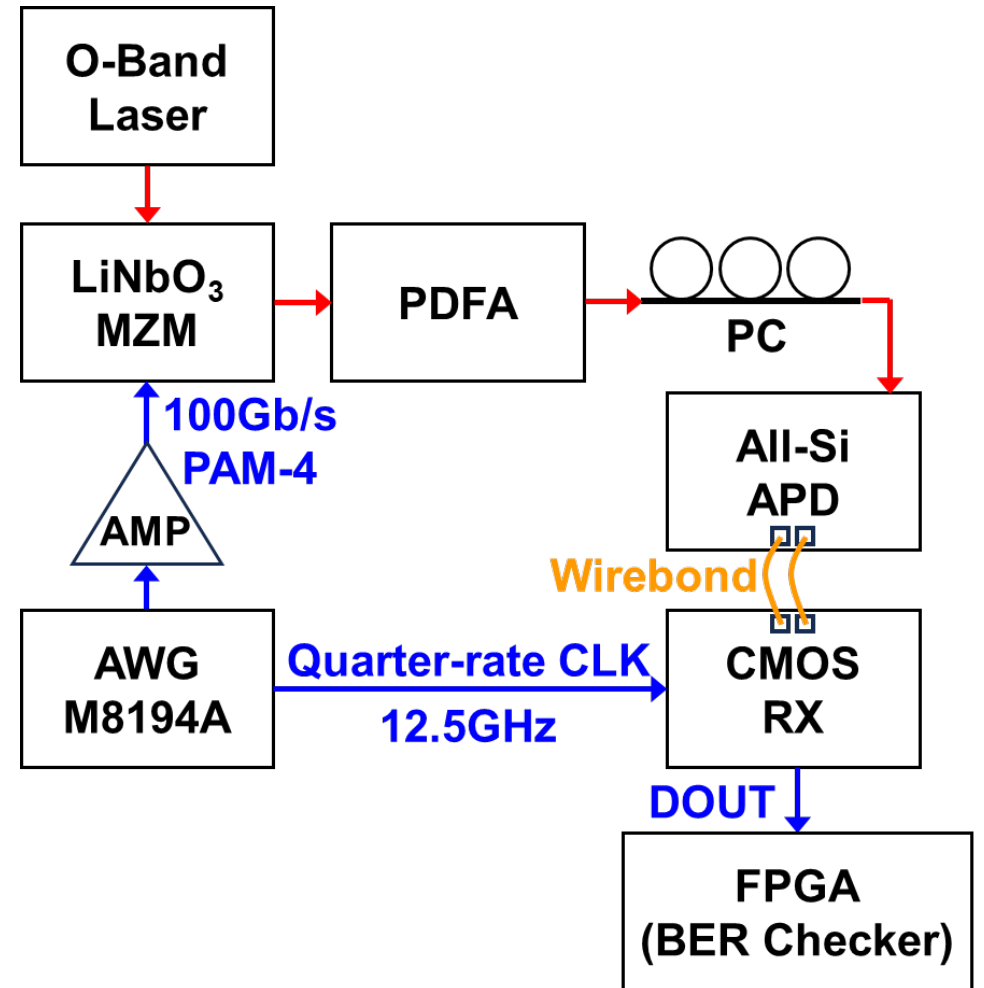
# All-Si APD and 28nm CMOS RX Integration

- All-Si APD and CMOS RX front-end integrated using wirebonds
- Vertical fiber coupling using grating couplers
- Heaters on PIC utilized to tune microring APD to drop optical signal to a given CMOS RX front-end channel



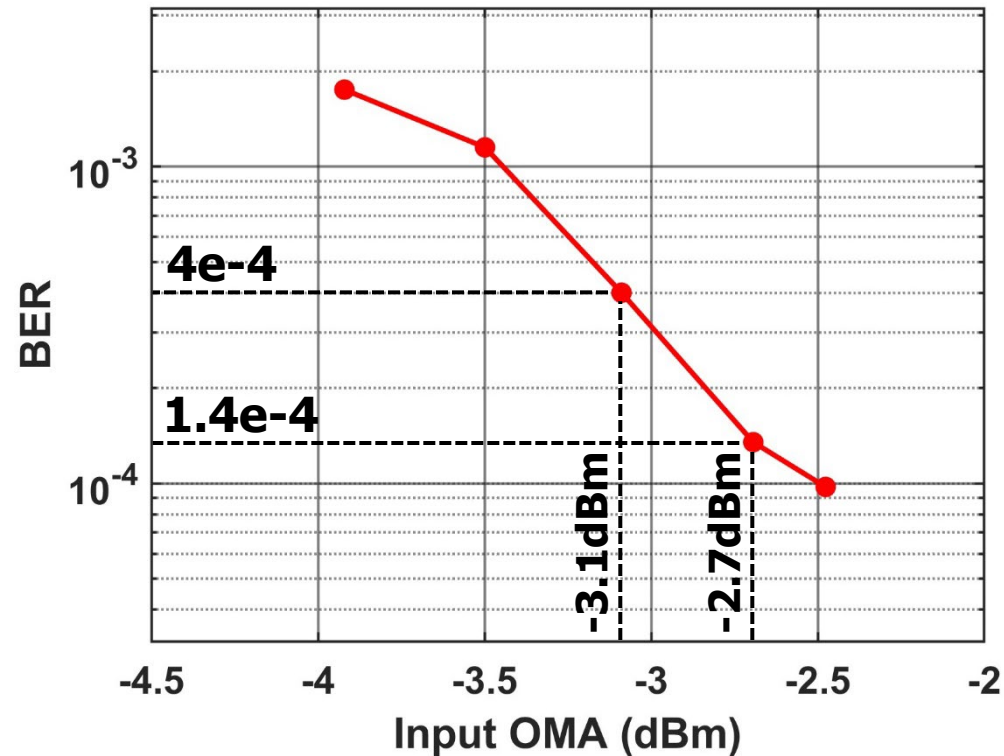
# Optical Testing Setup

- External O-band laser provides CW light to the MZM
- AWG generates electrical 100Gb/s PAM-4 signal and 12.5GHz quarter-rate clocks to the CMOS RX
- 100Gb/s PAM-4 optical output of the MZM is fed to the All-Si APD through the PDFA and grating coupler
- Measured deserialized MSB/LSB bits for BER measurement

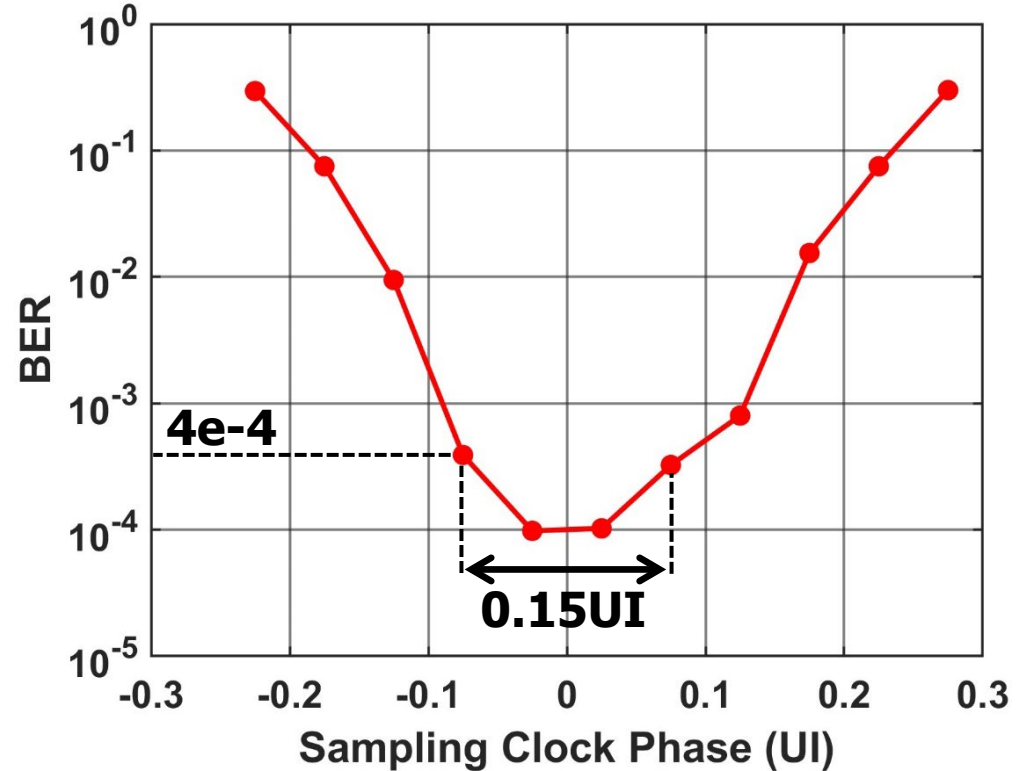


# RX Sensitivity and Bathtub Measurements

OMA Sensitivity at 100Gb/s PAM-4



100Gb/s BER with -2.48dBm OMA

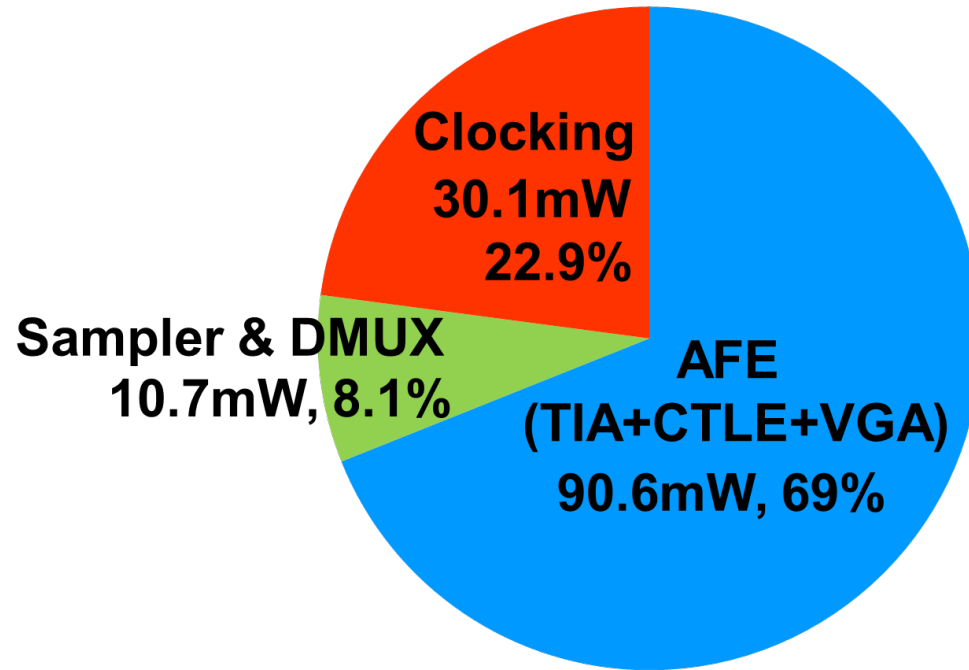


- Measured with PRBS-9 pattern
- -2.7dBm sensitivity at  $1.4e-4$  BER

# Power Breakdown

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**RX Total Power**  
**131.4mW at 100Gb/s PAM-4**



- 1.3pJ/b energy efficiency
- AFE power consumption is dominant



# Receiver Performance Comparison

|                             | E. Sentieri<br>ISSCC 2020 | R. Liu<br>IPC 2023 | H. Li<br>JSSC 2022 | This Work          |
|-----------------------------|---------------------------|--------------------|--------------------|--------------------|
| CMOS Technology             | 55nm BiCMOS               | 28nm CMOS          | 28nm CMOS          | 28nm CMOS          |
| PD Material                 | Not reported              | Si-Ge              | III–V              | <b>All-Silicon</b> |
| Data Rate (Gb/s)            | 53.125                    | 56                 | 100                | 100                |
| OMA Sensitivity (dBm)       | -9.5                      | -7                 | -8.9               | -2.7               |
| BER                         | 2.4e-4                    | 1e-4               | 2.4e-4             | 1.4e-4             |
| PD Responsivity (A/W)       | 0.945                     | 1.7                | 1                  | 0.29               |
| RX Energy Efficiency (pJ/b) | 4.4                       | 1.61               | 3.9                | <b>1.3</b>         |

- Best energy efficiency among slicer-based PAM-4 RX
- First demonstration of optical receiver integrating All-Si APD with CMOS front-end

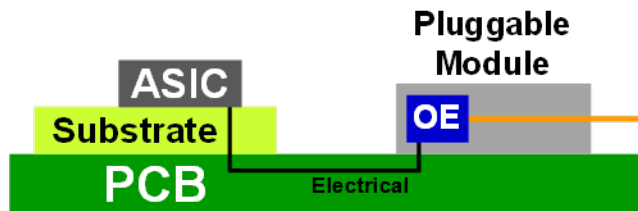
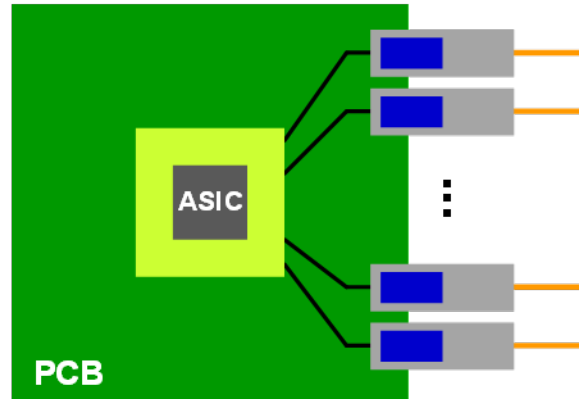
# Outline

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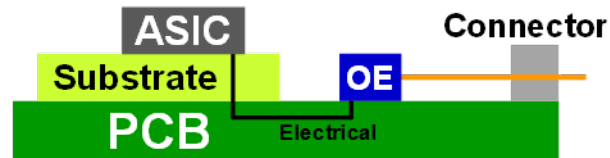
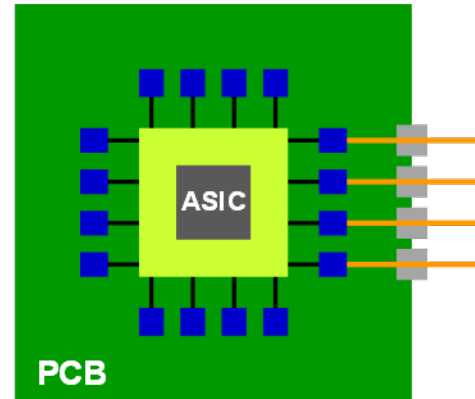
- Motivation
- 80Gb/s PAM4 Optical DAC Microring Modulator Transmitter
- 100Gb/s PAM4 Optical Receiver with All-Silicon Microring APD
- 100Gb/s 1.32pJ/b PAM4 Optical Receiver with Digital CDR
- Conclusion

# Optical I/O

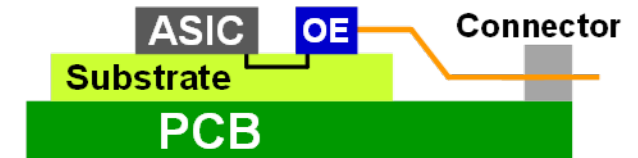
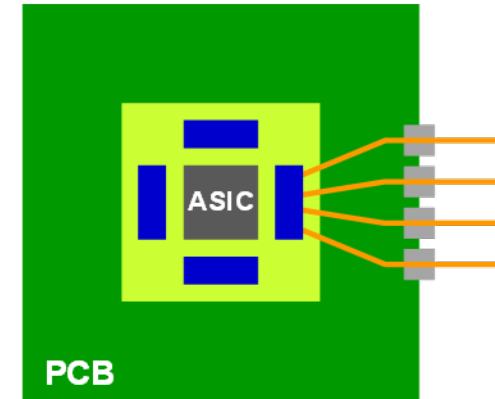
## Pluggable Optics



## On Board Optics



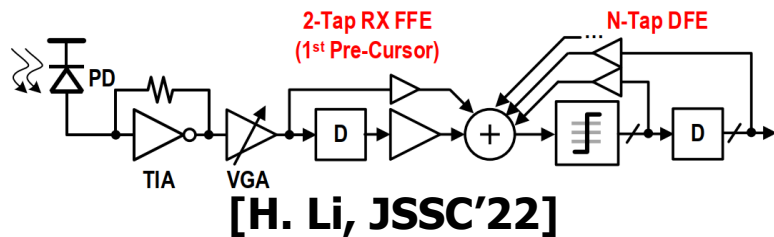
## Co-Packed Optics



- Optical I/O overcomes the physical and scaling limitations of electrical links
- Optical interconnects are shifting from board-level links to in-package integration, bringing PIC closer to the EIC

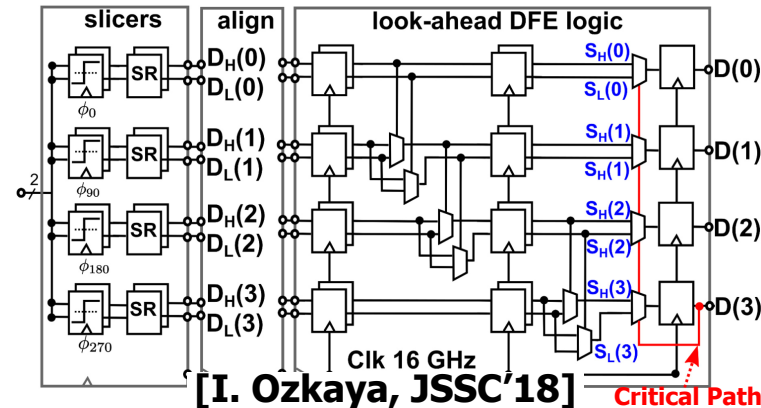
# Prior Art: Decision-Feedback Equalizer

## Direct-feedback DFE



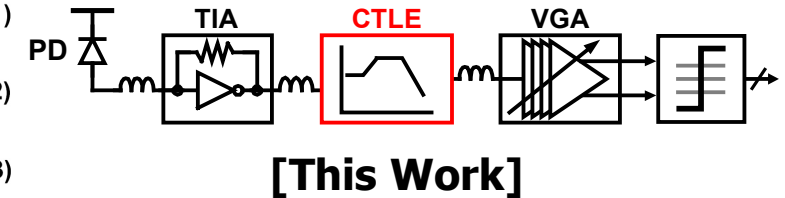
- 😊 Removes residual ISI without amplifying noise
- 😞 Requires S&H and integrating summer to avoid conventional 1UI latency: System complexity and power consumption↑

## Speculative DFE



- 😊 Relaxes AFE BW requirement
- 😊 No 1UI loop latency
- 😞 System complexity ↑: Multiple threshold for PAM-4
- 😞 Increased AFE loading
- 😞 More stringent AFE linearity requirement

## CTLE

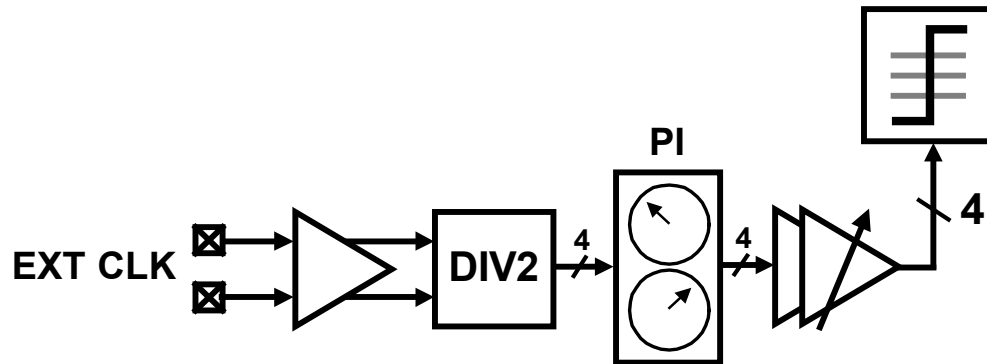


- 😊 System complexity and power consumption
- 😊 No 1UI loop latency
- 😞 Requires Higher AFE BW
- 😞 Cannot fully remove residual ISI

- ❑ CTLE is chosen for reduced system complexity and power consumption
- ❑ To minimize residual ISI penalty, sub-Nyquist TIA BW with BW recovering CTLE is implemented

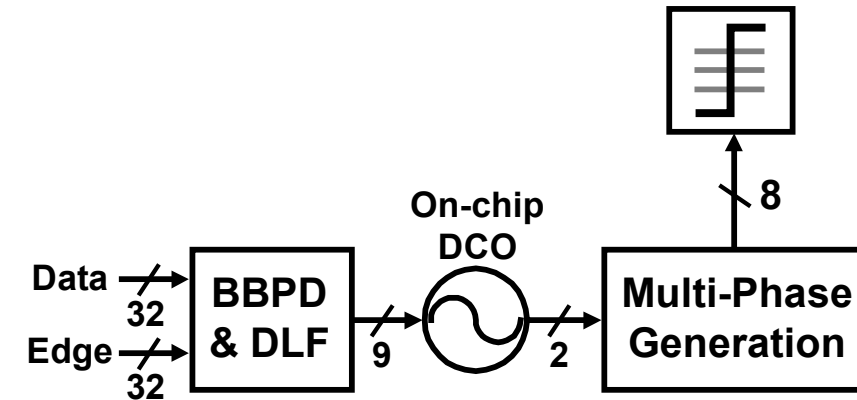
# Clocking Architecture

## PI-based CDR



[H. Li, JSSC'22]

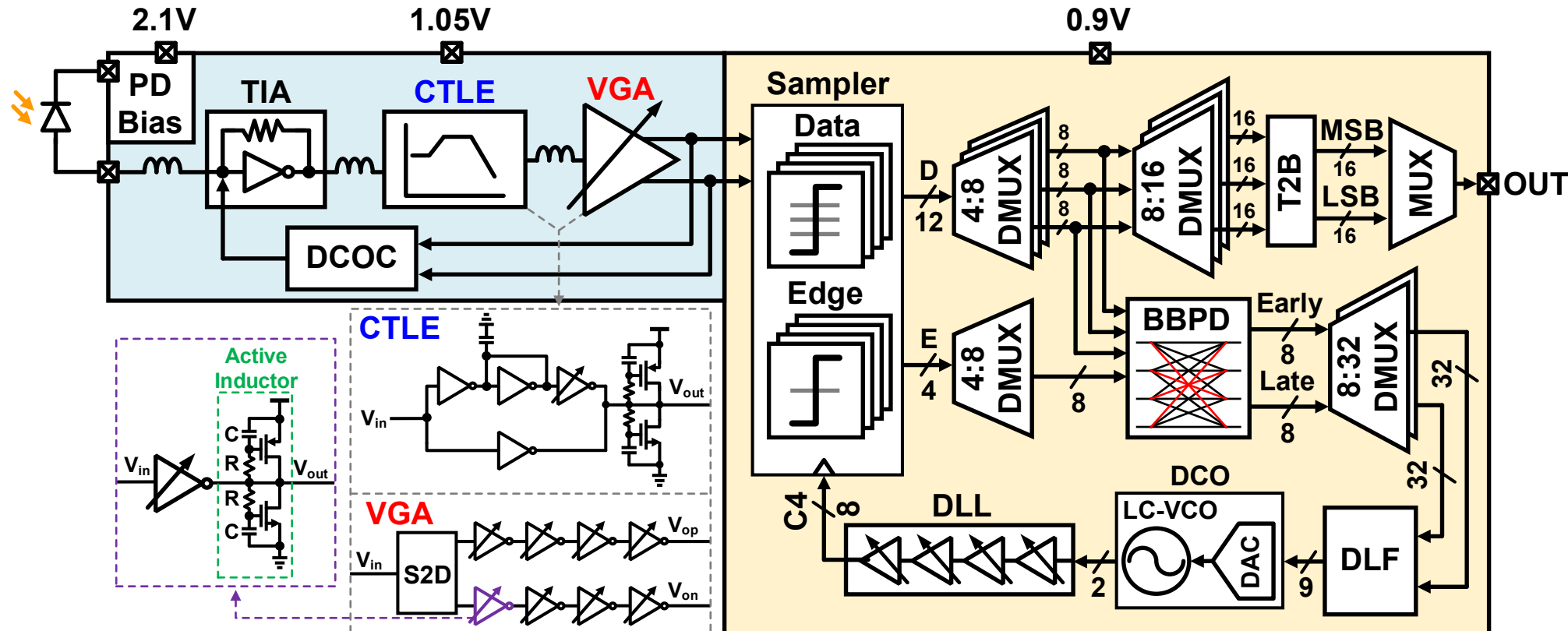
## On-chip Clock Generation CDR



[This Work]

- 😊 Allows for shared PLL across multi-channel
  - 😞 Needs external clock distribution network and phase-interpolator → Increase system complexity and power consumption
  - 😊 Simple clock distribution
  - 😞 Requires VCO → More challenging design
- This work combines CTLE-based RX equalization with on-chip clock generation to reduce system complexity and power consumption

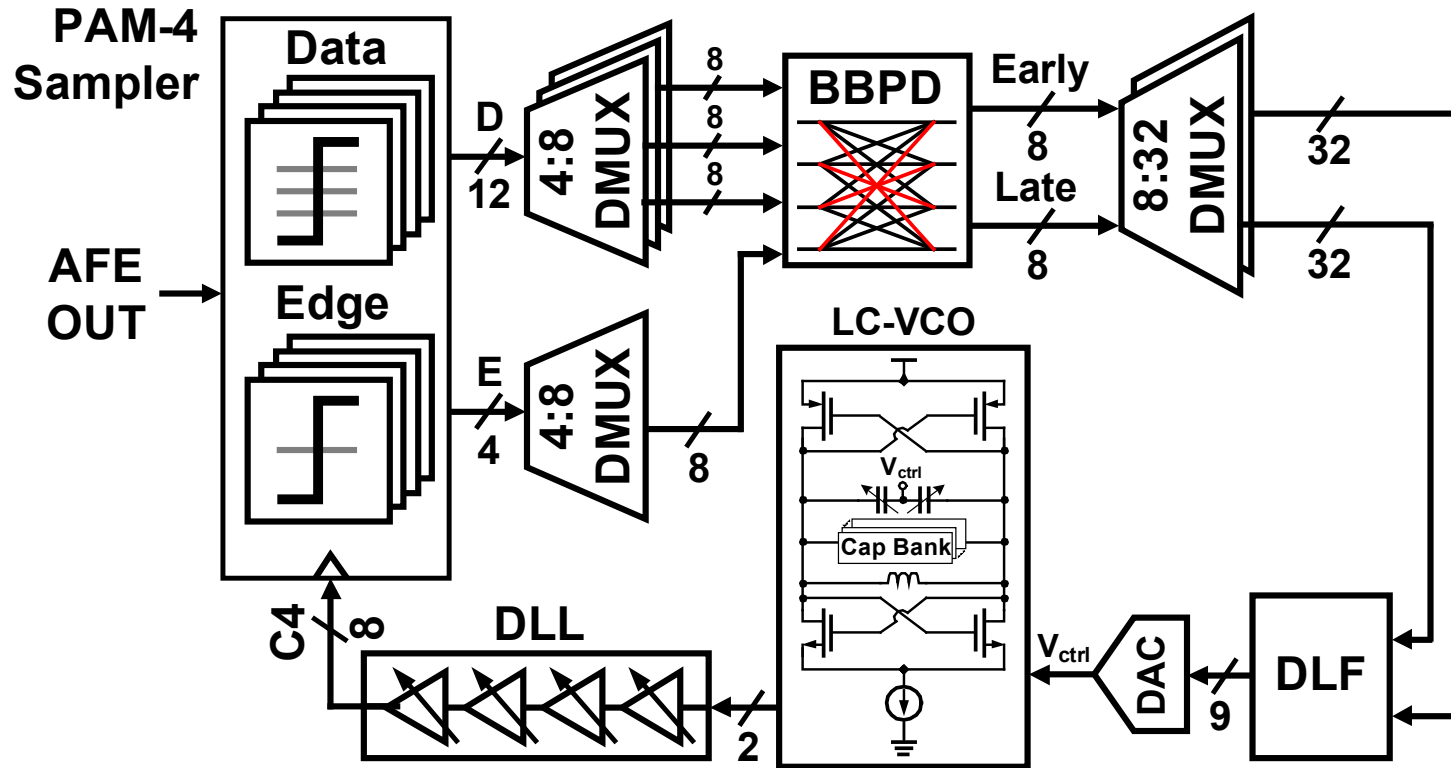
# Optical Receiver Architecture



- ❑ PD integrated with CMOS RX through wire-bonding
- ❑ Low-noise AFE by combining sub-Nyquist TIA with BW recovering CTLE/VGA
- ❑ Digital CDR with LC-DCO supports on-chip quarter-rate clock generation



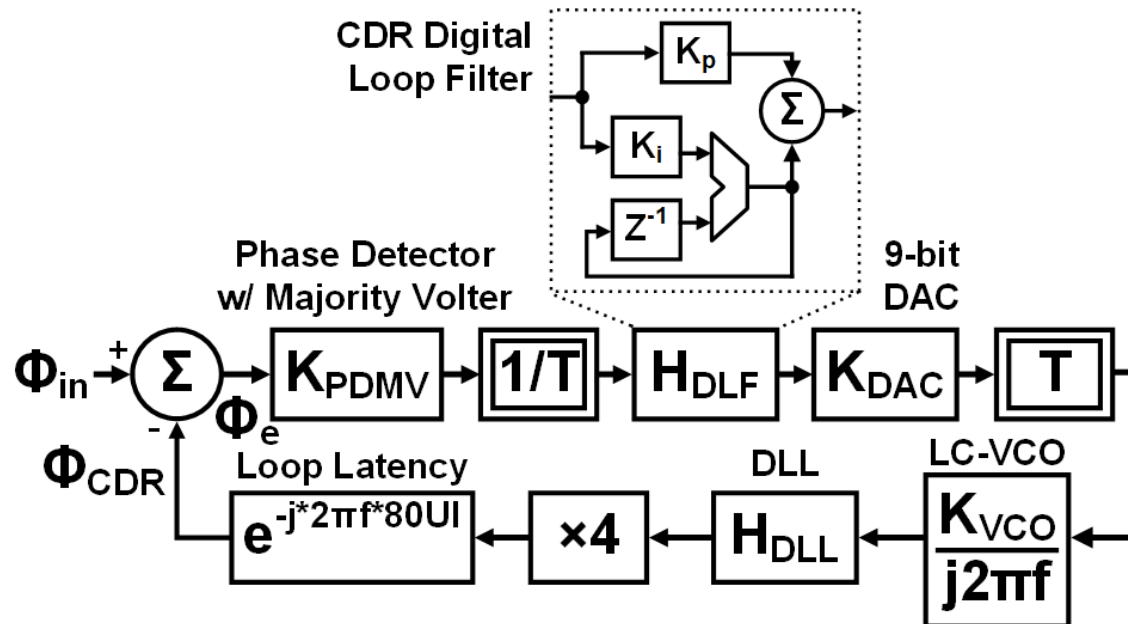
# Digital CDR with LC-DCO



- PLL-based digital CDR generates quarter-rate differential clocks
- DLL generates 8-phase clock signals for  $2\times$  oversampling
- DLF with proportional ( $K_p$ ) and integral ( $K_i$ ) gain control

# CDR Frequency Domain Model

## CDR Freq. Domain Model



## CDR Closed-loop Transfer Function

$$TF(f) = \frac{LG(f)}{1 + LF(f)},$$

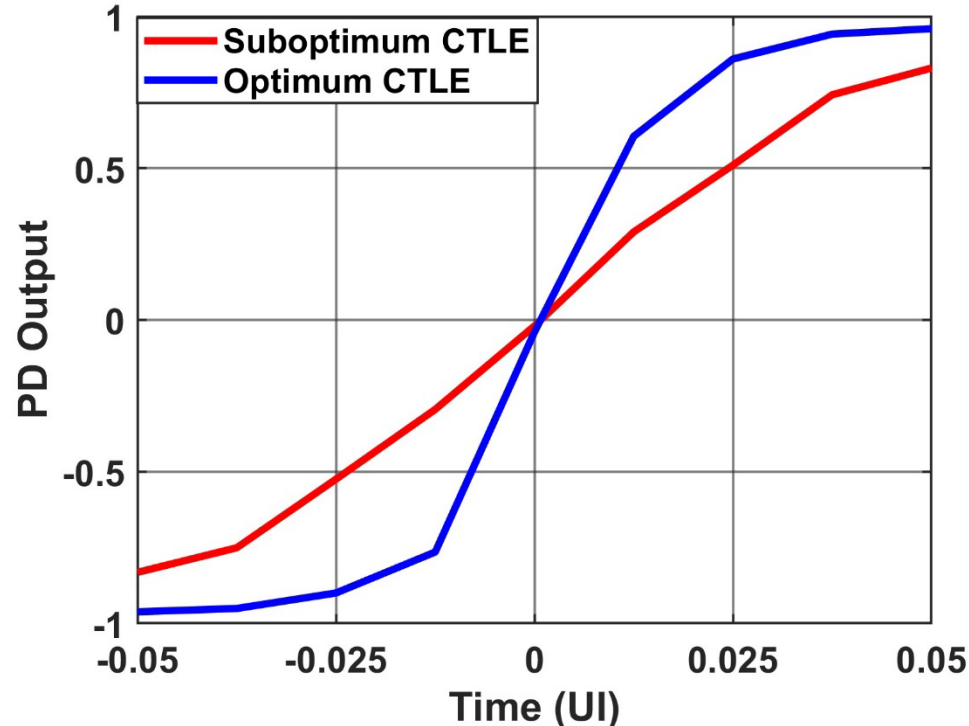
$$LG(f) = 4(K_{PDMV})H(e^{j2\pi fT})(K_{DAC})\left(\frac{K_{VCO}}{j2\pi f}\right)(H_{DLL}(f))(e^{-j2\pi f*80UI})$$

## CDR Parameters

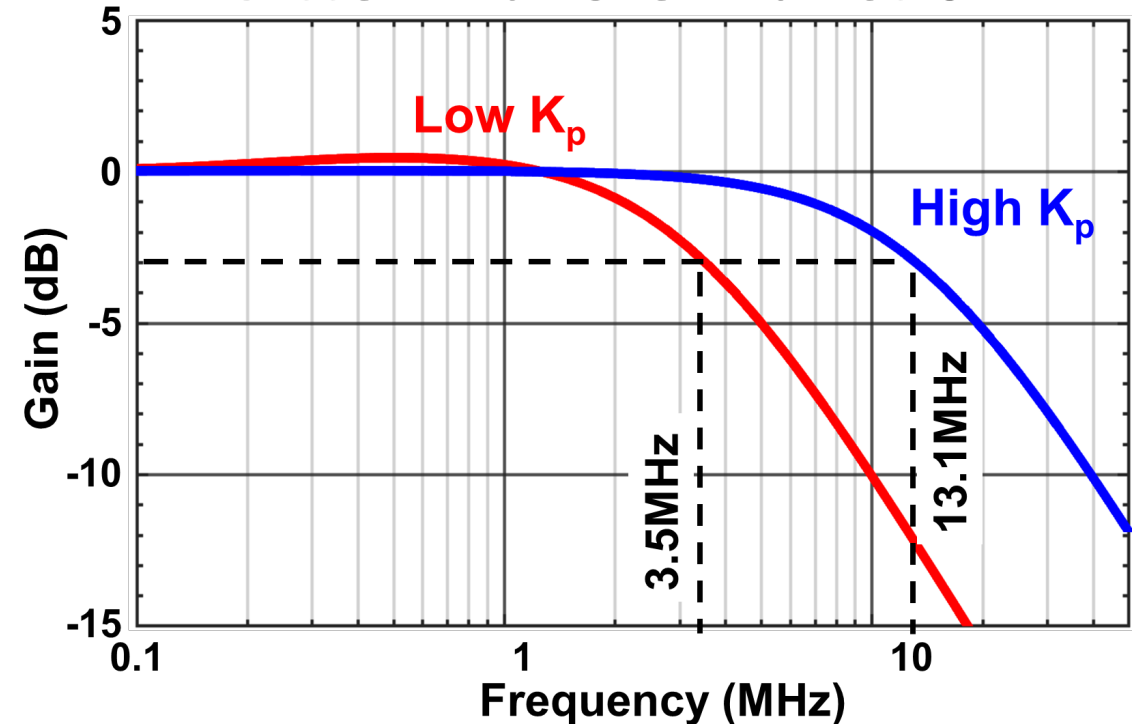
| Param.     | Value     | Unit   |
|------------|-----------|--------|
| $K_{PDMV}$ | 3.3       | Per UI |
| $K_p$      | 2         |        |
| $K_i$      | $2^{-14}$ |        |
| $K_{DAC}$  | $0.4/2^9$ | V/step |
| $K_{VCO}$  | 750       | MHz/V  |

# Transfer Function of Digital CDR

## BBPD Transfer Function

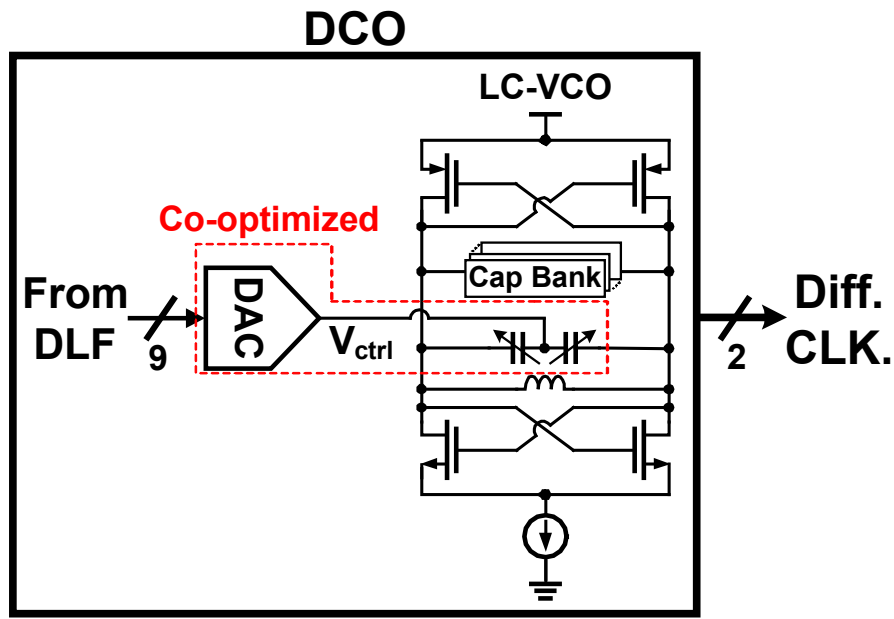


## Jitter Transfer Function

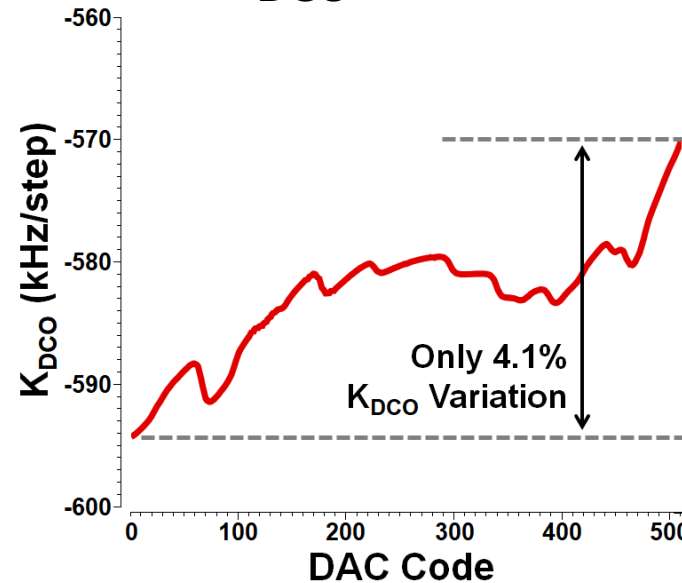


- BBPD gain reduces with sub-optimum CTLE setting due to increased ISI
- DLF programmability allows CDR BW and stability optimization

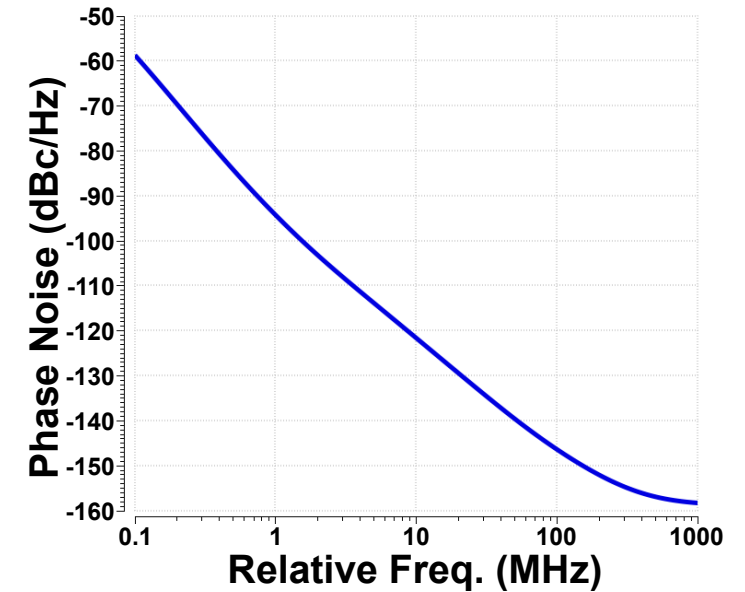
# DCO (Digitally-Controlled Oscillator)



**$K_{DCO}$  Curve**

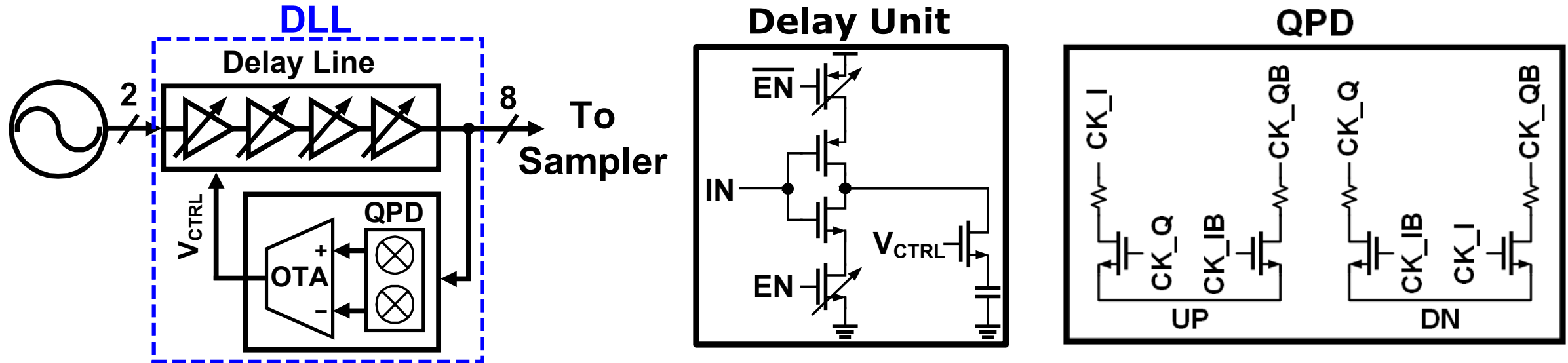


**DCO Phase Noise**



- DCO consists of 9-bit DAC and LC-VCO
- DAC and oscillator co-optimized for linear frequency tuning
- LC-Oscillator selected for lower phase noise

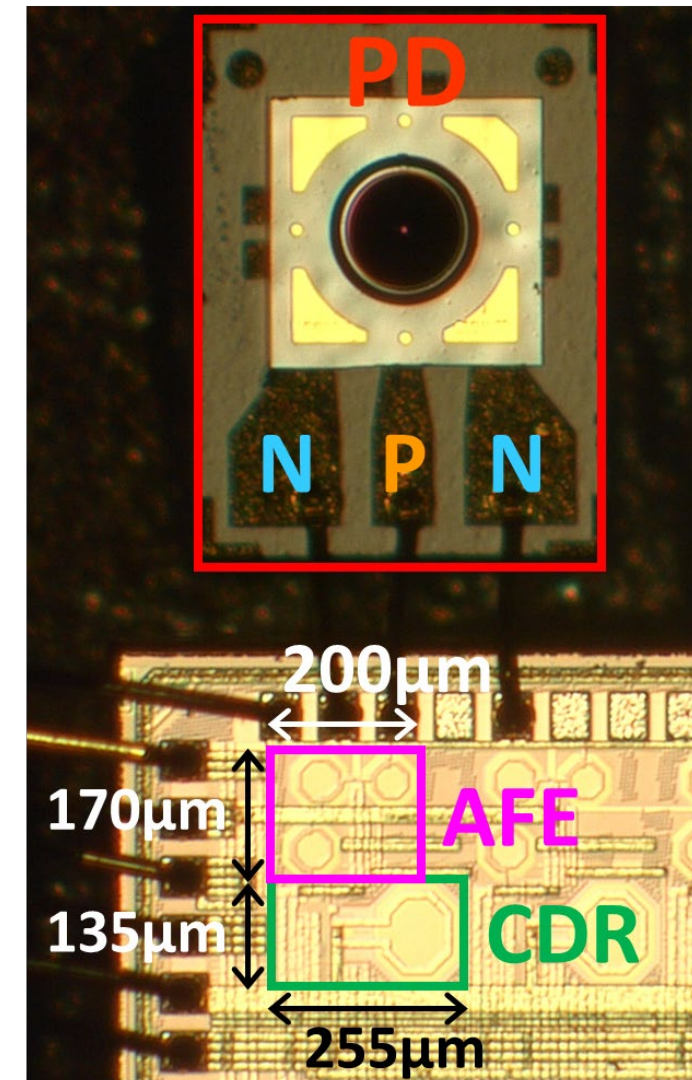
# DLL for Multi-Phase Generation



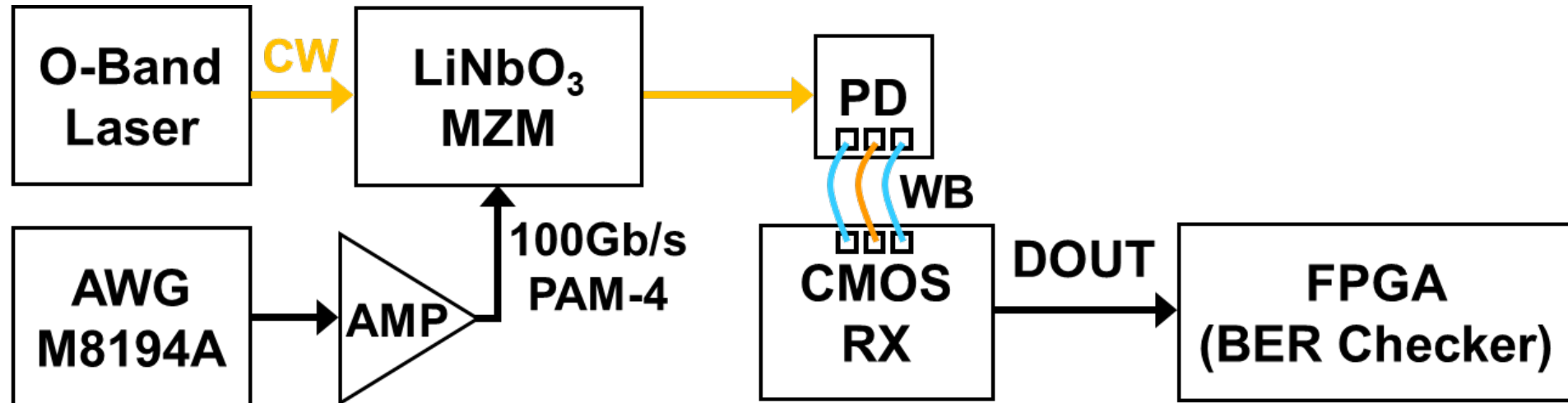
- DLL generates 8 phases from the differential VCO clock for quarter-rate, 2× oversampling architecture
- Negative feedback to control delay with load capacitance
- DLL BW is significantly higher than CDR BW to avoid interference

# PD – CMOS RX Integration

- Commercial III–V p-i-n PD
- CMOS RX chip
  - 28nm bulk process
- Integrated using wire-bonding
- Lens-assisted optical coupling



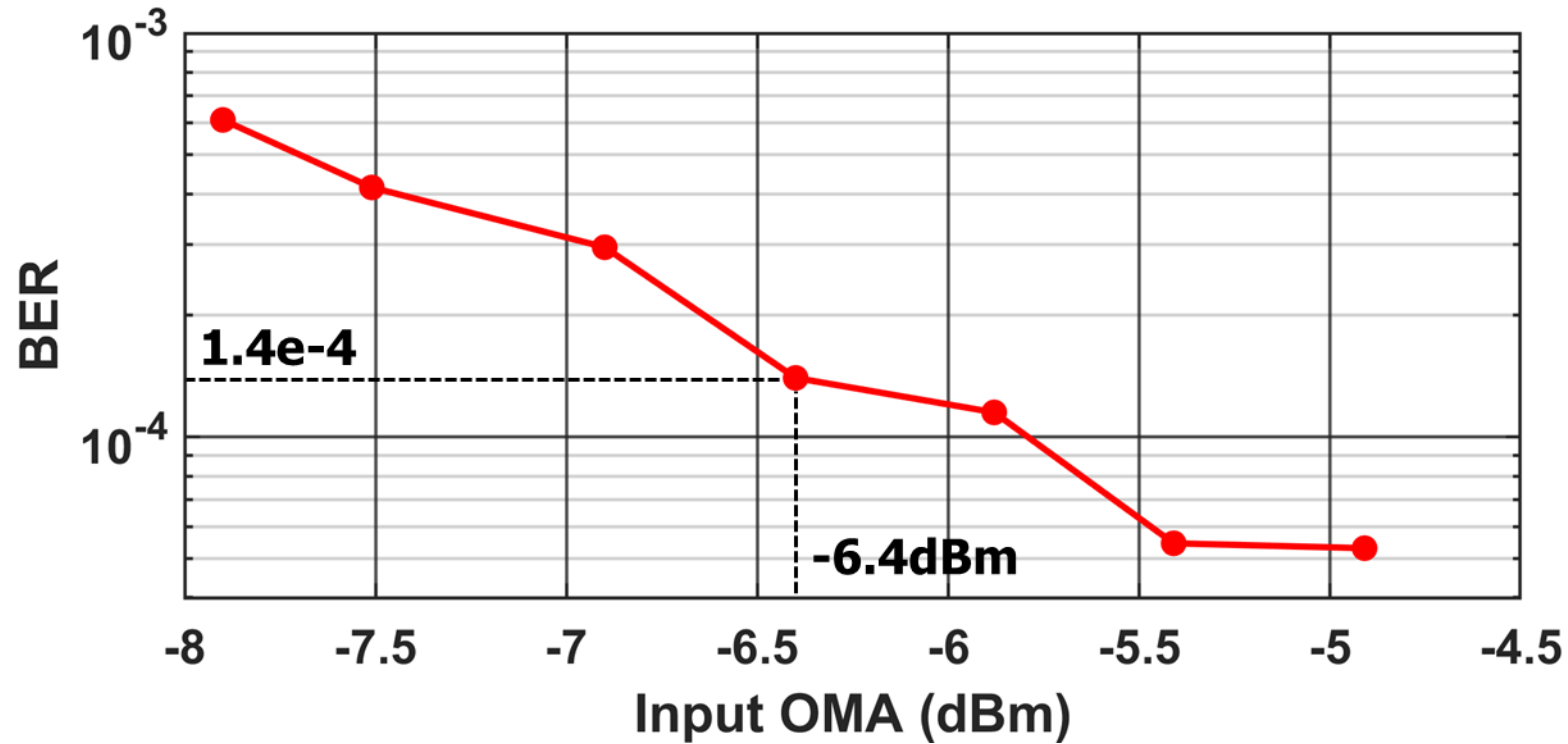
# Optical Testing Setup



- External O-band laser provides CW light to the MZM
- AWG generates electrical 100Gb/s PAM-4 signal
- 100Gb/s optical output of the MZM is fed to PD through lens
- Measured deserialized MSB/LSB bits for BER measurement

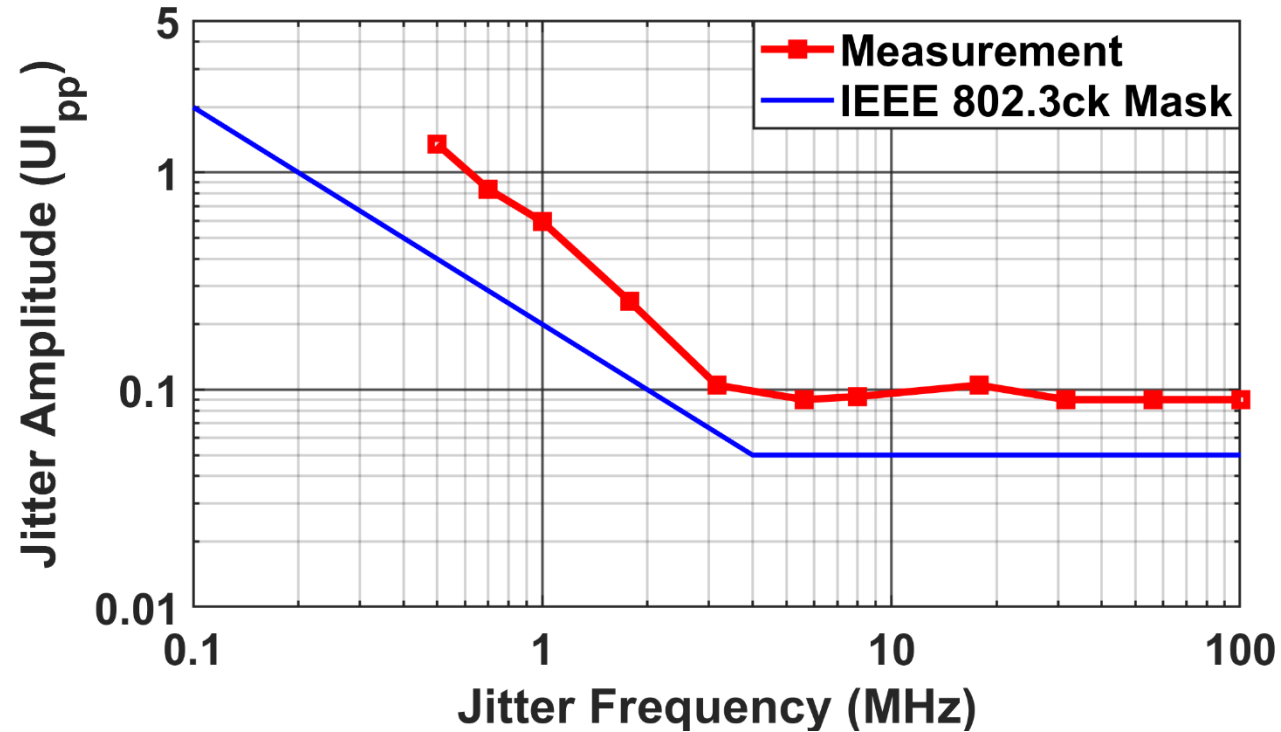


# 100-Gb/s PAM-4 OMA Sensitivity



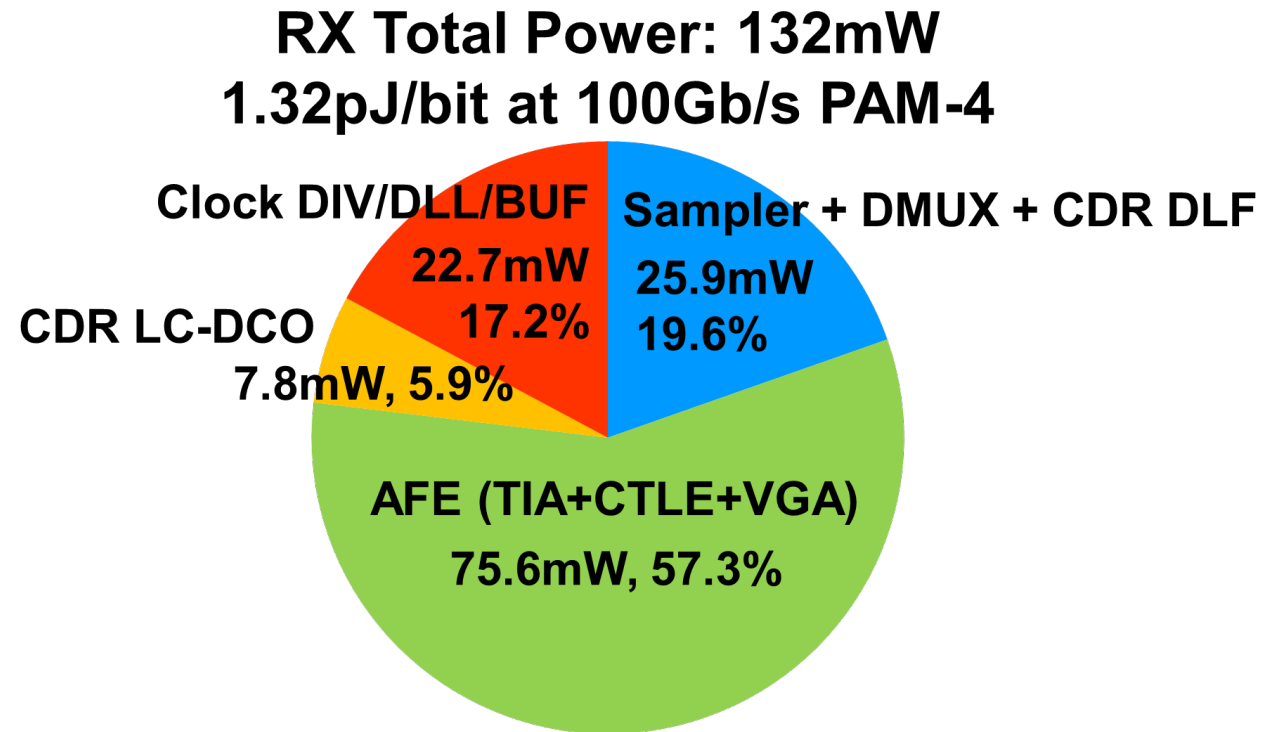
- PRBS-15 pattern with 3.68 dB extinction ratio
- -6.4dBm sensitivity at  $1.4 \times 10^{-4}$  BER

# Jitter Tolerance



- JTOL measured with the input OMA of -5.4 dBm ( $\text{BER}=1.4 \times 10^{-4}$ )
- CDR bandwidth around 4 MHz with 0.09UI high-frequency JTOL
- Satisfies the IEEE 802.3ck mask

# Power Breakdown



- CTLE based EQ. saves power consumption compared to DFE based EQ.
- On-chip CDR with DCO removes the need for external clocking, leading to power efficiency improvement

# Performance Comparison

|                             | E. Sentieri<br>ISSCC 2020 | I. Ozkaya<br>JSSC 2018 | H. Li<br>JSSC 2022   | This Work            |
|-----------------------------|---------------------------|------------------------|----------------------|----------------------|
| Technology                  | 55nm BiCMOS               | 14nm FinFET            | 28nm CMOS            | 28nm CMOS            |
| Data Rate (Gb/s)            | 53.125                    | 60                     | 100                  | <b>100</b>           |
| Modulation                  | PAM-4                     | PAM-2                  | PAM-4                | PAM-4                |
| On-Chip CDR?                | Yes                       | Yes                    | No                   | <b>Yes</b>           |
| OMA Sensitivity (dBm)       | -9.5                      | -8.2                   | -8.9                 | -6.4                 |
| BER                         | $2.4 \times 10^{-4}$      | $1 \times 10^{-12}$    | $2.4 \times 10^{-4}$ | $1.4 \times 10^{-4}$ |
| PD Responsivity (A/W)       | 0.945                     | 0.52                   | 1                    | 0.65                 |
| RX Equalization             | N/A                       | Lookahead DFE          | FFE + DFE            | CTLE                 |
| RX Energy Efficiency (pJ/b) | 4.4                       | 1.9                    | 3.9                  | <b>1.32</b>          |

□ Best energy efficiency among slicer-based optical RX

# Conclusion

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- Energy-efficient CMOS PAM4 transceiver front-ends are necessary to support future co-packaged optics
- Optical DAC-based transmitter heterogeneously-integrated with a two-segment microring modulator and 28nm CMOS MSB/LSB drivers that utilize independent swing control and asymmetric FFE achieves 80Gb/s operation
- 100Gb/s 28nm CMOS PAM4 receiver uses all-Si APD and low-bandwidth TIA followed by CTLE to improve OMA sensitivity
- Digital CDR with LC-DCO supports on-chip clocking, leading to lower power and reduced system complexity